ISSN No.IS-2454-8782

Lac, Plant Resins and Gums Statistics 2018: At a Glance



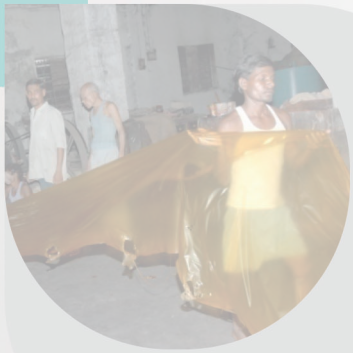




ICAR-INDIAN INSTITUTE OF NATURAL RESINS AND GUMS
Namkum, Ranchi- 834 010 (Jharkhand)

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Bulletin (Technical) No: 1/2021

Correct citation

Yogi RK, Kumar Nirmal and Sharma KK 2021. Lac, Plant Resins and Gums Statistics 2018: At a Glance. ICAR-Indian Institute of Natural Resins and Gums, Ranchi (Jharkhand), India. Bulletin (Technical) No. 1/2021. 1-80 pp.

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Row 2– Natural pine resin
Row 3 – Grading activities in gum industry

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ISSN No.IS-2454-8782
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FOREWORD

Non Wood Forest Products (NWFPs) based raw materials are supplied for food, paint, varnish, cosmetic and pharmaceutical industries over the world. Natural resins and gums are export oriented commodities and mostly exported in raw form. It is essential to process and add value to these for enhanced domestic consumption and also to create employment opportunities for the local populace. India is one of the largest producers of natural resins, gums and gum-resins (NRGs) along with China, Indonesia, Russia and Brazil. Our country is the world leader in production of *guar*, gum *karaya* and psyllium gums as well as lac.

The eastern region of India is lagging in agriculture and is home to more than 50 % of India's poor and food insecure population. Lac cultivation is an important vocation for inhabitants of 45 out of 150 disadvantaged districts of India. Jharkhand - the leading state in lac production (55% of National production). About 25-32% of total farm income in major lac growing areas comes from lac cultivation. It is a major source of livelihood and income generation especially for tribal farmers in rain-fed areas. About one million man-days are generated in the existing lac processing units alone. Utilizing full potential of available host plants for lac cultivation with recommended technologies may be promising strategy to check the migration and providing the opportunities for employment at local level. Subsequently, it will enhance the income level and by thus standard of living of the tribal farmers. Government of India has initiated the Minimum Support Price (MSP) for 25 NWFPs as marketing strategy on the top priority to boost up the sector.

The present publication, 'Lac, Plant Resins and Gums Statistics 2018: At a Glance' contains statistics on geographical distribution, production, price, processing, export and import of natural resins and gums including lac, pine resins, *guar* gum, gum *karaya*, *dhawda* gum, *psyllium* gum, tamarind gum powder, *Cassia tora*, *Olibanum*, *etc.* The production of gums in important states, *viz*, Chhattisgarh, Madhya Pradesh, Gujarat, Rajasthan and Andhra Pradesh; production of pine resins in Uttarakhand and Himachal Pradesh; export and import of natural resins and gums is included to give a clear and comprehensive picture of the importance of these at national and international level.

In this issue, content is presented in five major headings covering introduction; methodology; production, processing and trade; policy implications and conclusions. Information on market trend is essential for better organization of Indian producing states to meet the demand through better organization of their local/regional/national commercial channels from production to export. Further, stabilization of the market with appropriate stocks, quality control of exported products at global level and adequate remunerative price support at production level in India is also covered. In this issue, data pertaining the comparison of XI and XII plan periods is highlighted.

The statistics presented in this publication offer valuable insights of NRG sector. I am sure that the information and data contained in this bulletin would be useful to all the stakeholders of NRG sector. Authors made every effort to distill and condense a very large and diverse topic into and approachable volume. My appreciation goes to authors for their efforts made in bringing out this bulletin. Suggestions and inputs are sought from stakeholders for improvement of this publication in future.

Ranchi

(KK Sharma)
Director

Acknowledgement

We thankfully acknowledge the International Trade Centre (ITC), a joint agency of the World Trade Organization and the United Nations, Genewa, Switzerland; Director General of Commercial Intelligence and Statistics (DGCIS), Kolkata, Government of India; Department of Industrial Policy and Promotion, Ministry of Commerce and Industry, New Delhi, Government of India; Shellac and Forest Products Export Promotion Council (SHEFXIL), Kolkata; Directorate of Economics and Statistics, Ministry of Agriculture and Farmers Welfare, Government of India; The Tribal Cooperative Marketing Development Federation of India Limited (TRIFED), Ministry of Tribal Affairs (MOTA), New Delhi, Government of India; Girijan Co-operative Corporation Limited (GCCL), Visakhapatnam, Government of Andhra Pradesh; Kovel Foundation, Visakhapatnam, Andhra Pradesh; Sahayog Community Coordination Network, Visakhapatnam, Andhra Pradesh; The Jharkhand State Co-operative Lac Marketing & Procurement Federation Ltd. (JASCOLAMPF), Ranchi, Government of Jharkhand; State Agriculture Marketing Board (JHAMFCOFED), Ranchi, Government of Jharkhand; Chhattisgarh Minor Forest Product Federation Ltd.(CGMFP), Raipur, Government of Chhattisgarh; Madhya Pradesh State Minor Forest Product Federation Ltd. (MPMFP), Bhopal, Government of Madhya Pradesh; Gujarat State Forest Development Corporation (GSFDC), Vadodara, Gujarat; Forest Department, Government of Himachal Pradesh; Forest Department, Government of Uttarakhand; Forest Department, Government of Maharashtra; various NRG based processors, traders, exporters, importers, progressive farmers, forest dwellers, officials of various Government and Non - Government Organizations (NGOs) and other key informants for their valuable inputs, information and data. Publications from various organizations of United Nations also referred for world level secondary data and other relevant observations.

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I. INTRODUCTION

The globalization and rapid economic integration in the changing global scenario is putting enormous pressure on the natural resources. Consequently the role of forests too has become very crucial. In the present scenario of global warming and its control, forests have attracted particular interest since they play an essential role in sequestering carbon from the atmosphere. Forest plays a crucial role in adaptation and mitigation to climate change. Forests help to store more carbon than any other terrestrial ecosystem (India State of Forest Report, 2019). In the Forest Report, 2019, the total carbon stock in forest is estimated as 7,124.6 million tons. There is an increase of 42.6 million tons in the carbon stock of the country as compared to the last assessment 2017. Lac cultivation requires plant hosts which act as carbon sink. Total carbon stock under *ber*, *palas* and *kusum* is 17552 kg/ha, 18816.93 kg/ha and 36573.55 kg/ha, respectively.

Forest sustains environmental goods and service particularly food and wood, maintain hydrological cycle with ground water recharge and combating with degradation of landscapes, disaster management and pollution abatement. Global agenda of delivering sustainable development and addressing climate change can be delivered only if all nations act upon their fair share of responsibilities including the fulfillment on means of implementation by the developed world to the developing countries. India has been continuously moving towards economic growth, keeping in mind the imperatives of sustainable development. India is among a few countries in the world where forest and tree cover have increased considerably. The forest and tree cover have reached 80.73 million hectare which is 24.56 % of the geographical area of the country (Economic Survey 2018-19, GOI). India accounts for 2 % of the total global forest area in 2015 as per the Global Forest Resource Assessment (FRA) by Food and Agriculture Organization (FAO). National Mission for a Green India envisages a holistic view of greening and focuses on multiple ecosystem services along with carbon sequestration and emission reduction. The mission emphasized the landscape approach to treat large contiguous areas of both forest and non-forest, public and private lands with a key role of the local communities in planning, implementation and monitoring.

According FAO, Non Wood Forest Products (NWFPs) are the goods of biological Origin other than wood derived from the forest. Though there is no official census figures for the forest dependent population in the country, different estimates put the figures from 275 million (World Bank, 2006) to 350- 400 million (MoEF, 2009). These NWFPs may be in the form of leaves, flowers, seeds, fruits, stems, tannin, gums and resins, herbs, medicinal plants, bamboos *etc.* which are the source of income to about 350 millions of people across the world (Mukul, *et al.* 2014; Bhat, 2012) and 50 millions of people in India (Bhat, 2012). In India, about 15000-18000 plant species were reported, out of which about 3000 species yield NWFPs (Murthy et al., 2005). However, only 126 species yielding NWFPs which have been commercialized in India (Maithani, 1994; Murthy *et al.*, 2005 and Yadav *et al.*, 2019c).

In past, as per the Forest Act 1927 lac secreted by a tiny insect *Kerria lacca* is defined as the forest product as collected by the forest dwellers. Subsequently the transformation of the rural structure, farmers started its cultivation scientifically on various lac host trees.

Recently a bushy plant is also introduced by the institute in various states. Consequently, more than 95% production is routed through the farming community and it is no more the collection by the forest department. Hence, a policy interventions need to be taken to declare it as an agricultural produce.

The share of agriculture and allied sectors in Gross Value Added (GVA) of the country at current prices declined from 18.2 % in 2014-15 to 17.2% in 2017-18. The share of forestry and loggings to the GVA of the country at current prices has also been declining and it is fluctuating between 1.58% in 2013-14 to 1.20 % in 2017-18. The growth of GVA of agriculture and allied sectors has witnessed a fluctuating trend. GVA at constant prices (2011-12) for 2017-18 from 'Agriculture, Forestry and Fishing' sector is estimated to grow by 5.0 % as compared to growth of 6.3% in 2016-17 (Economic Survey 2018-19, GOI).

According to ITC calculations based on UN COMTRADE statistics, the world trade aggregation of lac, natural gums, resins, gum-resins and balsams during 2017 was 1314.63 million US dollars. Out of this, the world export aggregation of lac, natural gums, resins, gum-resins and balsams during 2017 was about 552.39 million US dollars.

Share of NRGs in India's total import (₹25776.75 billion in 2016-17 & ₹30010.33 billion in 2017-18) has been declined slightly at 0.053% (₹13.72 billion) in 2016-17 and 0.049% (₹14.69 billion) in 2017-18. Although, the share of NRGs in India's total agri import increased from 0.88% in 2016-17 to 1.03 % in 2017-18. Share of NRGs in India's total export (₹18494.34 billion in 2016-17 & ₹19565.15 billion in 2017-18) has been increased from 0.17% (₹31.08 billion) in 2016-17 to 0.21 %(₹40.65 billion) in 2017-18 with an annual acceleration of 30.78 % (Department of Commerce, 2019). Similarly, the share of NRGs in India's total agri export has also been increased from 1.37% in 2016-17 to 1.62 % in 2017-18.

A decadal data (2008-2017) on world EXIM (Export–Import) aggregation of lac, natural gums, resins, gum-resins and balsams were analyzed and presented in Figure 1 and Figure 2. Since 2012, deceleration in the value of world export aggregation was observed, but it increased in 2017. Stagnation was found in the value of world import aggregation during the similar period.

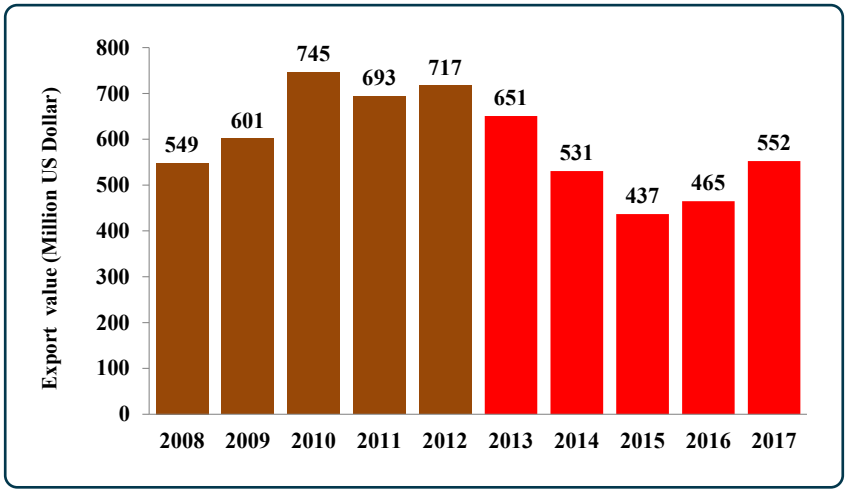


Figure 1. World export flow of lac, natural gums, resins, gum-resins and balsams

Major suppliers of NRGs are contributing about 92 % share in international market are France (24.2%), Sudan (20.8%), India (12.9%), Indonesia (8.0%), Germany (6.5%), USA (6.2%), Brazil (5.3%), UK (4.1%), Greece (2.2%) and Italy (1.9%). Rests of the 7.9% NRGs are supplied from 58 countries across the world (Figure 3). Similarly, the world import aggregation of lac, natural gums, resins, gum-resins and balsams during 2017 was about 756.63 million US dollars. Major importers of NRGs contributing about 80% share in the international market are India (22.7 %), France (13.0 %), United States of America (9.5 %), Germany (6.3 %), China (5.7 %), Spain (4.9%), Portugal (3.7%), United Kingdom (3.7 %), Italy (3.5 %), Russian Federation (2.9%), Netherlands (2.5 %), Ireland (2.5%), Switzerland (1.6%), Japan (1.5%), Bulgaria (1.2%) and Brazil (1.2%). During 2017, rests of the 13.6% demand of NRGs aroused from the 65 countries across the world (Figure 4).

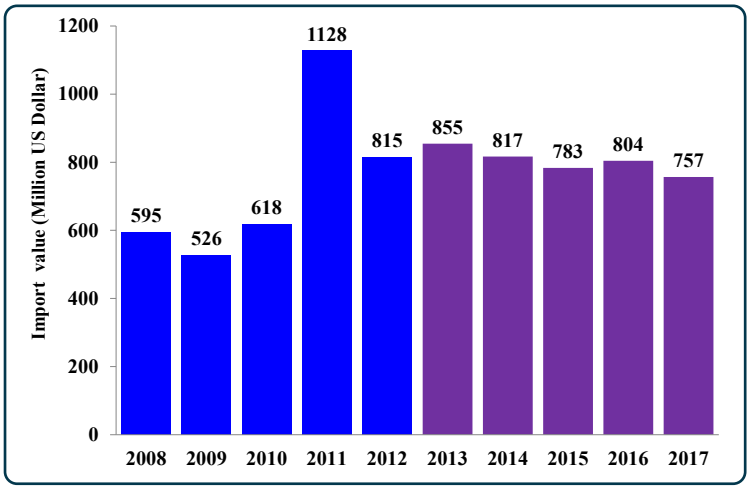


Figure 2. World import flow of lac, natural gums, resins, gum-resins and balsams

About 70% of the NWFP collection in India takes place in the tribal belt of the country (Mitchell *et al.,* 2003). Around 55% of employment in forestry sector is attributed to this sector alone (Joshi, 2003). NRGs of commercial importance like lac (*Kerria lacca* Kerr), pine resin (*P. roxburghii* Sarg.), *guar* gum (*Cyamopsis tetragonoloba* L.), gum *karaya* (*Sterculia urens* Roxb.), *dhawada* gum (*Anogeissus latifolia* Roxb.), tamarind gum (*Tamarindus indica* L.), *char /piyar* gum (*Buchanania lanzan* Spreng.) and *babool* gum (*Acacia nilotica* L.) are produced in India. India holds monopoly in international trade over some of the NRGs such as lac, gum *karaya, psyllium gum* and *guar* gum.

NWFPs based on their chemical composition may be classified in three categories namely natural resins, natural gums and gum resins. Natural resins are solid or semi-solid materials, usually a complex mixture of organic compounds called terpenoides, which are insoluble in water but soluble in certain organic solvents. Resins are secretion of several plants, particularly coniferous trees.

Resins are used in the production of varnishes, adhesives and food glazing agents. These are also used as raw material for synthesis of incense and perfume. This group of natural resins includes lac secreted by an insect *K. lacca* (Kerr) and plant originated products like rosin, copal and dammar. Solidified resin from which the volatile terpene components have been removed by distillation is known as rosin.

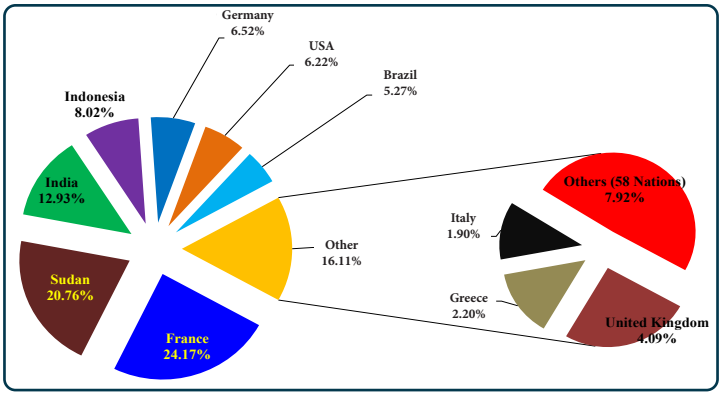


Figure 3. Break up of the World export aggregation of NRGs

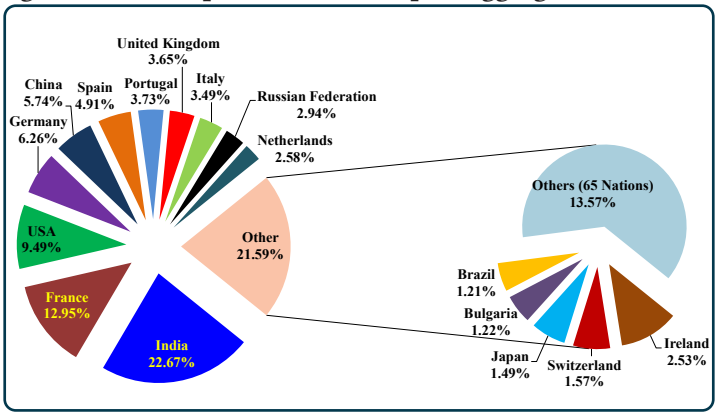


Figure 4. Breakup of the World import aggregation of NRGs

Table 1. World EXIM trade aggregation of lac, natural gums, resins, gum-resins and balsams

(Value in Million US$)

Year	Export	Import	Re-Export	Re-Import	Total Trade
2006	669.12	559.72	11.77	0.75	1241.35
2007	470.47	575.00	11.82	1.50	1058.79
2008	548.82	594.65	13.26	0.12	1156.86
2009	601.20	526.39	19.94	0.24	1147.76
2010	745.48	617.75	15.46	0.20	1378.89
2011	693.28	1127.70	15.12	0.55	1836.64
2012	716.89	814.98	17.88	0.24	1550.00
2013	650.94	854.70	17.06	0.31	1523.02
2014	530.78	816.86	10.43	0.37	1358.44
2015	436.70	783.45	5.87	0.58	1226.60
2016	465.03	804.37	9.06	0.20	1278.67
2017	552.39	756.63	5.42	0.19	1314.63

Source: ITC calculations based on UN COMTRADE statistics.

Natural gums are polysaccharides of natural origin, capable of causing a high viscosity in the solution. Most often these gums are found as exudates from woody elements of plants or in seed coatings. In the food industry these are used as thickening, gelling and emulsifying agents and stabilizers. These are also used as adhesives, binding agents, crystal inhibitors, clarifying agents, encapsulating agents, flocculating, foam stabilizers, swelling agents, *etc.* Natural gums can be classified according to their origin. Firstly, tree exudates *e.g.* gum Arabic, Asian gum, African gum, gum *ghatti*, gum *tragacanth*, gum *karaya*, *chicle* gum, dammar, mastic gum and spruce gum. Secondly, the seed based *viz. guar* gum, tamarind gum, *psyllium* gum, locust bean gum, cassia tora seed, *etc.* Thirdly, originated from seaweeds *e.g.* agar and *carrageenan* and forthly, produced by bacterial fermentation *e.g. gellan* gum and *xanthan* gum.

Gum-resins are the natural mixtures of gums and resins in variable proportions therefore possess properties of both the groups. They contain traces of essential oils and are partly soluble in water. They have a penetrating and characteristic odour and taste and obtained from the plants. Olibanum/*salai* gum (*Boswellia serrata*), guggal *(Commiphora wightii),* myrrh, asafoetida, *etc.* are the major gum resins of national importance.

II. METHODOLOGY

Reliable and periodical assessments of quantities and values of production and market outlook studies at the national level of NWFPs are essential to decision-makers for policy formulation and governing the sustainable development of the sector. Appropriate and biometrically valid inventories of non-wood forest resources are an essential prerequisite for their sustainable management and harvesting. Timely and accurate estimation of production may be helpful for the stakeholders to plan their operations in time. Accuracy in production estimate would be helpful in precision planning by all concerned. Besides knowing the present status, it would be helpful in regulating imports, planning for enhanced exports, reasonable prices and reliability in supply of lac based products. However, such statistics do not yet exist for most countries, neither is the already available data comparable among countries. In most countries, the current coverage and quality of existing information is inadequate for policy analysis and decision making at national level.

The objectives of present methodology are to estimate and update the production, processing and value addition of NRGs at national level and to compile the latest EXIM data. The approach used in the present methodology was through survey of local traders and processors, as all NRGs produced in India are collected/procured through the local traders/societies/forest departments. This report presents an overview of quantity and values for selected forest products and services from Indian forests. Results highlight a strong concentration of value in *guar* gum production. The economic value of NWFPs however is only partly reflected on the market and recorded by official statistics. Underestimation by existing statistics might be due to several reasons, including: (i) the public-good nature of many products/services and consequent difficulties in estimating them; (ii) the fact that data recorded for official statistics are not always complete (for example they do not always cover all the region as in the case of NWFPs); and (iii) the fact that a certain proportion of the NRGs is not mirrored by official data because some products are traded through informal channels and markets.

Traders/federations are limited in number but they have close contact with the primary purchaser who have knowledge of present crop condition and expected output. Survey of all big traders was helpful in estimation. Further, all the produced products pass through the processing units. Processors use the current harvested or stocked or imported crop/ produce. Survey of processing units was helpful in estimating quantity of processed products at national level and validation of production data. Information on market arrival during seasons at important markets in India was collected through survey of identified major markets. Regular contacts were also made with the persons/ organizations related to NRGs in India through correspondence, phone and personal visit for collection and updating of data. Production estimation was made by the survey of selected processors, exporters, importers and markets (traders). In the Indian context, the data relating to India's exports and imports by commodities is officially compiled and published by the DGCI&S, Government of India. The data is compiled in accordance with the Harmonized Commodity Description and Coding System, also known as the Harmonized System (HS) of tariff nomenclature, developed and maintained by the World Customs Organization (WCO) and adopted by the Government of India as Indian Trade Classification, commonly called ITC (HS) codes. Export and import data on NRGs & related products were obtained from Directorate General of Commercial

Intelligence and Statistics (DGCI&S), Kolkata as well as from the Department of Commerce, Ministry of Commerce and Industry. Secondary data were also collected from other agencies like Tribal Cooperative Marketing Development Federation of India (TRIFED), state forest departments & federations and Proprietor Datanet India Ltd. Five schedules / questionnaire were framed for collection of data and information during the field surveys *i.e.* survey of markets, survey of processing centers, crop-wise arrival in the market, survey of importers and processing at processing centers. Production year was considered from April to March (Financial year) and final production assessment completed by the month of April. Data and other related information for estimation of production and processing was collected throughout the year. Tabular analysis and pictorial presentation has been used to analyze the survey data. Validation of production data at national level was made on the basis of secondary data on quantity processed at national level, export and import figures.

Sampling design and survey area

Surveys were made in various NRGs producing areas of the country for collection of data throughout the year during 2017-18. National level information and data on NRGs were collected from primary and secondary sources. The requisite data were collected from respondents at various NRGs markets and processing centers based at throughout the potential states of the country. For updating the information and data, regular telephonic contacts were also made with the respondents.

Surveys were conducted in 56 districts of more than 20 states of India, Bangladesh and Nepal covering 1091 stakeholders including 467 institutions (Govt. Officials/ NGOs/ Other key informants), 368 farmers, 118 market functionaries (traders/ wholesalers), 138 processors/ Manufacturer /exporters/importers and resource persons were interacted through visits and telephonic conversations during 2017-18. These respondents were directly/indirectly concerned with NRGs production, processing, value addition across the country and overseas trade.

Name of the states and districts which were covered under survey and sample size surveyed during the year have been presented in Table 2 and Table 3, respectively. Category wise and state-wise composition of selected stakeholders is presented in Figure 5 and Figure 6.

Table 2. Details of the areas covered under the surveys during 2017-18

States/Country	Districts
Andhra Pradesh	Vishakhapatnam and East Godavari
Assam	Kamrup and Dhemaji
Bangladesh	Chapainawabganj
Bihar	Muzaffarpur and East Champaran
Chhattisgarh	Balarampur, Bilaspur, Dhamtari, Korba, Kanker, Korea and Raipur
Delhi	Chandni Chok
Gujarat	Bharuch, Surat and Vadodara
Himachal Pradesh	Sirmaur

States/Country	Districts
Jharkhand	Bokaro, Dhanbad, Giridih, Gumla, Khunti, Ranchi, Simdega and West Singhbhum
Karnataka	Bangaluru and Uttara Kannada
Madhya Pradesh	Balaghat and Seoni
Maharashtra	Bhandara and Gondia
Nepal	Chitwan and Kathmandu
NE region	Garo hills, Imphal, East Kameng, Barapani and Kohima
Odisha	Balasore and Sundergarh
Rajasthan	Ajmer, Bhilwara, Jaipur, Jodhpur and Udaipur
Tamin Nadu	Chennai & Coimbatore
Uttar Pradesh	Allahabad
Uttarakhand	Dehradun
Telangana	Hyderabad
West Bengal	Kolkata and Purulia

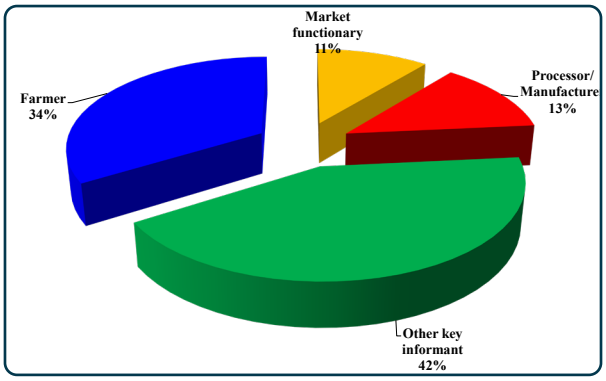


Figure 5. Composition of stakeholders under survey program (Category wise)

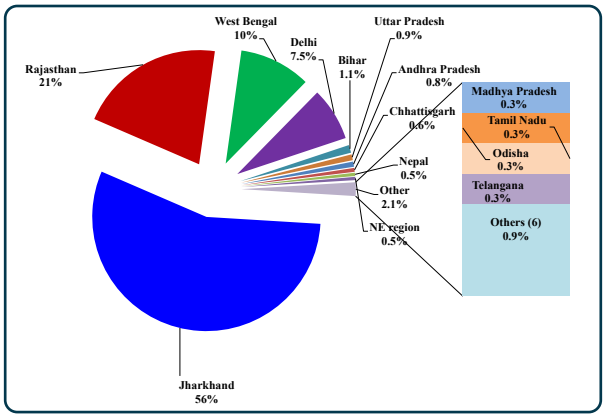


Figure 6. Composition of stakeholders under survey program (State-wise)

Table 3. Distribution of the sample size during the survey in 2017-18

States/ Country	No. of districts	Farmer	Market function-ary	Processor/ Manufac-turer / Exporters	Govt. Official/ NGOs/ Other key informant	Total
Andhra Pradesh	2	0	0	0	8	8
Assam	2	0	0	0	2	2
Bangladesh	1	0	0	1	1	2
Bihar	2	0	0	0	12	12
Chhattisgarh	7	0	0	0	7	7
Delhi	1	0	20	2	60	82
Gujarat	3	0	0	0	3	3
Jharkhand	8	315	34	47	207	603
Karnataka	2	2	0	0	0	2
Madhya Pradesh	2	0	0	0	3	3
Maharashtra	2	0	0	2	0	2
Nepal	2	0	0	0	5	5
NE region	5	0	0	0	5	5
Odisha	2	0	0	0	3	3
Rajasthan	5	41	40	58	86	225
Tamil Nadu	2	0	0	0	3	3
Telangana	1	0	0	0	3	3
Uttar Pradesh	4	0	0	0	9	9
Uttarakhand	1	0	0	0	2	2
West Bengal	2	10	24	28	48	110
Total	**56**	**368**	**118**	**138**	**467**	**1091**

III. PRODUCTION, PROCESSING AND TRADE IN NRGs

NRG production level during 2017-18 was estimated comparatively lower (5, 66,229 tons) than previous year (8, 43,178 tons). Production and trade of NRGs including *guar* gum, lac, pine resin, *karaya* gum, *dhawda* gum and other natural resins and gums is depicted in Table 4 and Figure 7.

Table 4. Total NRG production and trade during 2017-18 (quantity in tons)

Name of product	Production	% share	Export	% share	Import	% share
Guar gum*	638877.33	96.71	366664.17	96.49	95.21	0.09
Lac	14230.00	2.15	6889.48	1.82	300.00	0.28
Pine resin	5254.40	0.80	288.42	0.08	46703.63	43.49
Karaya gum	150.00	0.02	313.40	0.08	2299.98	2.14
Other NRGs	2100.00	0.32	5824.30	1.53	58000.74	54.00
Total	**660611.73**	**100.00**	**379979.77**	**100.00**	**107399.56**	**100.00**

**Estimation of guar gum is based on the conversion of total guar seed production*

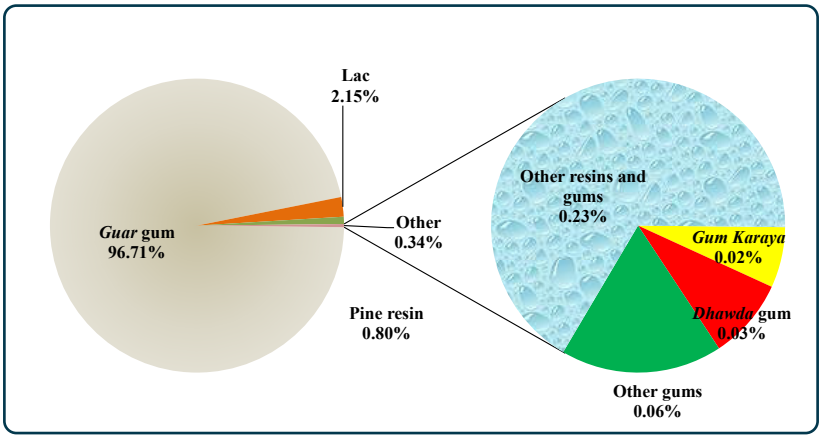


Figure 7. Product/major group-wise production share of NRGs

It is evident from the Table 4 that *guar* gum has a lion's share in total NRG production and consequently in export also. For *guar* gum, India is the leading producer, processor and exporter country in the world. Hence, a negligible quantity has been imported for research and other purposes. Similarly, India is the largest producer, processor and exporter of the lac. Export quantity of lac has been decreased by 4.85% in comparison to previous year export. However, the value of exported lac was increased by 9.18%. Only 3.29% production of the total NRGs is contributed by lac, pine resins, gum *karaya* and other NRGs while rest 96.71% is *guar* gum only.

Export under gum *karaya* and other NRGs has exceeded the production data as the exported quantity supplied from the carry over stock and secondly, raw material imported and

exported after processing in India. Overall, there was about 13.3% increase in total quantity of NRGs exported and quantity in import basket was declined by 2.3% over the previous year. But, higher price of *guar* gum in international market resulted with an increase of 28.5% in total foreign exchange earnings from the NRGs in comparison to 2016-17. The detailed information about state-wise production processing and trade of natural resins (Lac, pine resin, copal & dammer batu), natural gums (Gum *karaya*, *guar* gum, Cassia gum, *Psyliium* gum, tamarind gum, gum *ghatti, gum arabi, Asain gum, African gum, Tragacanth, Xanthan gum& other minor gums*) and natural gum resins (Olibanum, myrrh, asafetida & Dikamali) is presented below.

A. NATURAL RESINS

Lac – a fauna based natural resin



Photo 1. Natural Resin

Kerria lacca (Kerr), an Indian lac insect which thrives on the tender twigs of specific host trees *viz.*, *palas* (*Butea monosperma*), *ber* (*Ziziphus mauritiana*), *kusum* (*Schleichera oleosa*), *Flemingia semialata*, *Ficus spp. etc.* secretes resin (known as Lac). It is cultivated mainly by the tribal communities and a little quantity is collected by the forest dwellers in India. *Rangeeni* and *kusmi* are the two strains of lac insect which are classified based on preference of the insect for specific host plants. Raw lac is the source of three valuable products *i.e.* resin, dye and wax. Lac cultivation is an important source of income for livelihood of the forest and sub-forest dwellers of Jharkhand, Chhattisgarh, Madhya Pradesh, West Bengal, Maharashtra, Odisha and parts of Uttar Pradesh, Andhra Pradesh, Gujarat and NEH region. Lac production is highly labor intensive process and provides employment to both men and women dwelling in forest and sub-forest areas of these states. It is a highly remunerative crop, paying high economic returns to the farmers and also to foreign exchange of the country through its export. Lac is mainly produced in India, Thailand, Indonesia, parts of China, *etc.* India is the largest producer of lac in the world.

Lac production in India

Tree species in agro forestry is very important and vital for lac cultivation. It reflects the choice of farmers and in some sense demand of market. There are 49.2 crore of *palas* (*Butea monosperma),* 4.5 crore of *ber* (*Ziziphus mauritiana)* and about 3.7 crore of *Kusum (Schleichera oleosa)* stems spread in various Agro forestry Systems over the country **(FSI, 2017)** which is about 3% of total number of available trees. Current lac production estimates revealed that only less than 10% lac host trees are being utilized for lac cultivation. It is assumed that there are some bottlenecks hampering the growth of the lac sector. Estimation of lac production is required by the Government, lac-based industries, lac traders, entrepreneurs and exporters. The cultivation of lac on a large number of hosts of different kinds, its collection by numerous small growers, variations in the yield depending on the type and size of the host, cultivation practices and climatic conditions are the major factors influencing

the estimation of lac production. Production estimates of this commodity would be helpful in planning like overseas as well as domestic trade regulations, prices fixation and reliability in supply of lac based products.

On the basis of survey in the local weekly markets of different lac producing districts, the estimated national production of sticklac during 2017-18 was approximately 14,230 tons comprising *rangeeni* (5484 tons) and *kusmi* (8746 tons) sticklac. Among the lac growing states, Jharkhand state ranks 1st (54.6%) followed by Chhattisgarh (16.5%), Madhya Pradesh (13.9%), Maharashtra (6.0%), West Bengal (5.4%) and Odisha (2.2%). These six states contribute more than 98 % of the total lac production in India (Figure 8). Among to different cropping season crops, *aghani* crop was ranked 1st with the contribution of 34.4 % followed by *jethwi* (27.1 %), *baisakhi* (18.8 %) and *katki* (19.7 %) in total lac production.

Table 5. Lac production in India during 2017-18 (in tons)*

Name of states / Districts	Name of lac crop/strain@						Total (% share)
	Baisakhi	*Katki*	*Rangeeni*	*Jethwi*	*Aghani*	*Kusmi*	
Jharkhand	363	277	640	3226	3880	7106	**7746** (54.6)
Chhattisgarh	650	497	1147	420	685	1105	**2252** (16.5)
Madhya Pradesh	1187.5	951	2139	56	41	97	**2236** (13.9)
Maharashtra	390	465	855	2	8	10	**865** (6.0)
West Bengal	45	590	635	29	130	159	**794** (5.4)
Odisha	12.5	9	21	111	147	258	**279** (2.2)
Gujarat	10	2	12	5	5	10	**22** (0.6)
Uttar Pradesh	10	2	12	0	0	0	**12** (0.54)
Andhra Pradesh	5	3	8	1	0	1	**9** (0.12)
Assam	5	4	9	0	0	0	**9** (0.09)
Meghalaya	5	1	6	0	0	0	**6** (0.13)
Total	**2683 (18.85)**	**2801 (19.68)**	**5484 (38.53)**	**3850 (27.06)**	**4896 (34.41)**	**8746 (61.47)**	**14230 (100.00)**

*@**Baisakhi:** - Summer season crop of rangeeni; **Jethwi:** - Summer season crop of kusmi; **Katki:** - Rainy season crop of rangeeni; **Aghani:** - Winter season crop of kusmi;$ Andhra Pradesh including Telengana;* See* Annexure I *for details.*

In comparison to previous year, the production of *ranggeni* as well as *kusmi* crop during 2017-18 decreased by 11.4% and 13.9%, respectively. The decrease in production was comprised of 14.3 % and 13.5 % fall in the production of *jethwi and aghani* crops, respectively. However, a decrease of 11.5% and 11.4% was observed in case of *baisakhi* and *katki* crops, respectively .Overall, the total lac production for the year 2017-18 has been decreased by about 13 % in comparison to 2016-17. Lac production scenario in India during 2017-18 is presented in Table 5 and the top ten lac producing districts in the country are enlisted in Table 6. Share

of different crops at national level is presented in Figure 9. Lac production in India during previous five years is depicted in Figure 10.

Table 6. Top ten lac producing districts in the country

District (States)	2016-17	Rank	2017-18	Rank
Ranchi (Jharkhand)	2710	1	2540	1
Simdega (Jharkhand)	2115	2	1855	2
Khunti (Jharkhand)	1315	4	1115	3
Gumla (Jharkhand)	1336	3	995	4
Seoni (Madhya Pradesh)	865	5	864	5
Balaghat (Madhya Pradesh)	802	6	746	6
West Singhbhum (Jharkhand)	790	7	670	7
Gondia (Maharashtra)	655	9	660	8
Korba (Chhattisgarh)	705	8	630	9
Kanker (Chhattisgarh)	470	10	410	10

The overall production of the lac in the country has been estimated as 14,230 tons which is lower than the previous year (2016-17) production (16,352 tons). It is interesting to mention that the production level of lac is sustained to 14,230 tons during 2017-18 which was about 57 % higher in comparison to the lowest level of 9,035 tons during 2010-11. Although, it is about 32% lower than the highest level of 21,008 tons in 2013-14. District wise lac production statistics and major lac producing areas in the country have been presented in *Annexure I* and *Annexure II*, respectively.

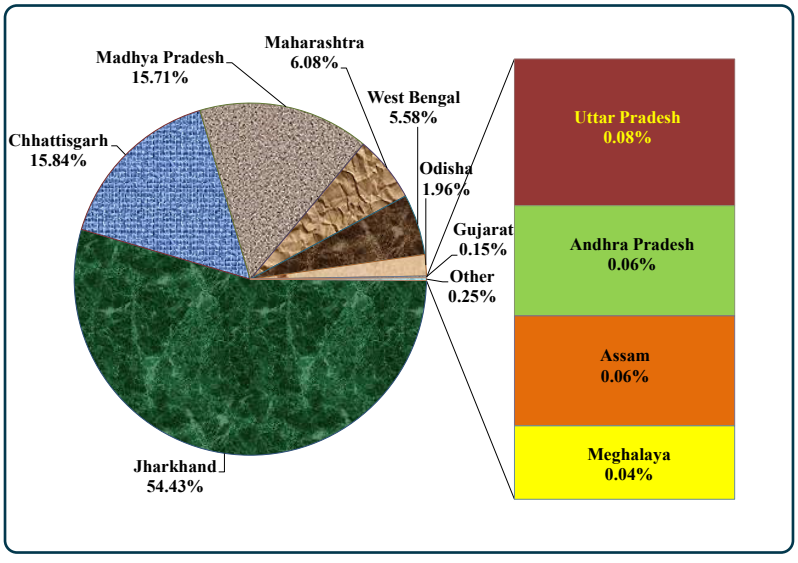


Figure 8. State-wise share in lac production during 2017-18

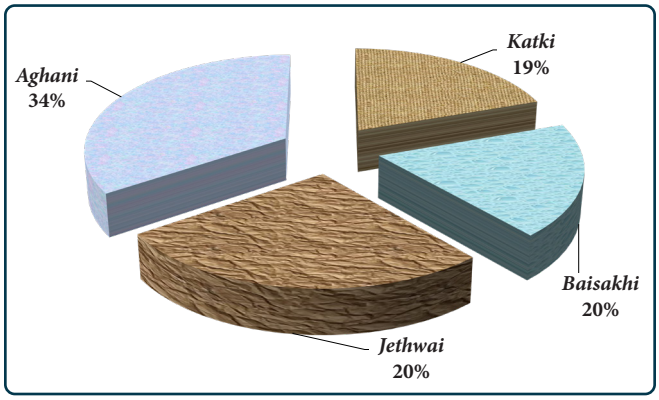


Figure 9. Crop-wise share in total lac production during 2017-18

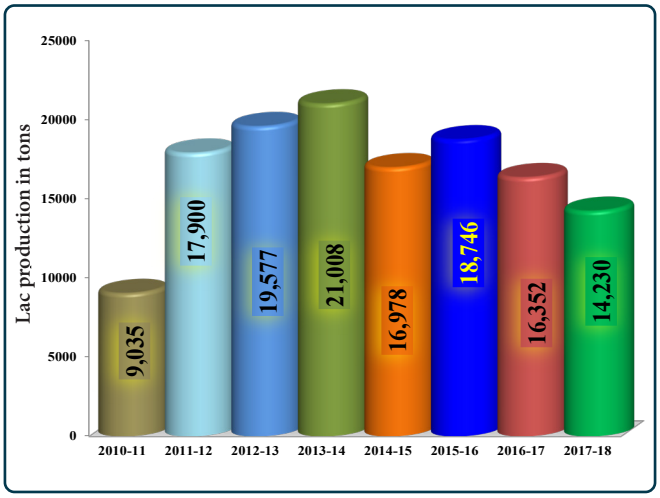


Figure 10. Lac production in India during previous years

Lac processing and value addition in India

Lac growers sell the sticklac or scraped lac in small quantity or lots in the rural weekly markets (*haats*). Rural markets (*haat*) in remote lac growing areas operate once or twice in a week. Lac growers, after harvesting sticklac sell to *paikars* (primary purchasers). Progressive lac growers, nearer to lac processing units sell their produce directly to processers. Subsequently, collection of scraped lac in the course of the market day, the *paikars* sell it to the wholesaler in the same market or nearby processing centers in bigger lots. Finally the wholesalers also sell their purchase to processor. Further, processed and value added products of lac supplied to domestic as well overseas firms for various industrial applications. Different lac processing centers in the country and amount of sticklac processed during 2017-18 have been presented in Table 7 and Table 8, respectively.

On the basis of surveys conducted at different lac processing centers in the country, the total quantity of sticklac processed during 2017-18 was 16156 tons which also included the imported as well as previous years carry over stock in India. Information about share of

different states in lac processing is presented in Figure 11 and amount of lac processed in India during previous five years is depicted in Figure 12. A total 174 lac processing units were functional in West Bengal (84), Chhattisgarh (31), Jharkhand (15), Maharashtra (6) and Madhya Pradesh (6) during the year 2017-18. In the processing of lac across the country, West Bengal (34.2 %), Jharkhand (25.3 %) and Chhattisgarh (22.4 %), shared about 82 % followed by Maharashtra (10 %), Madhya Pradesh (4.5%) and others (3.2%). There were 9 primary and 6 secondary markets existing at national level, in which annual arrival of sticklac was more than 500 tons.

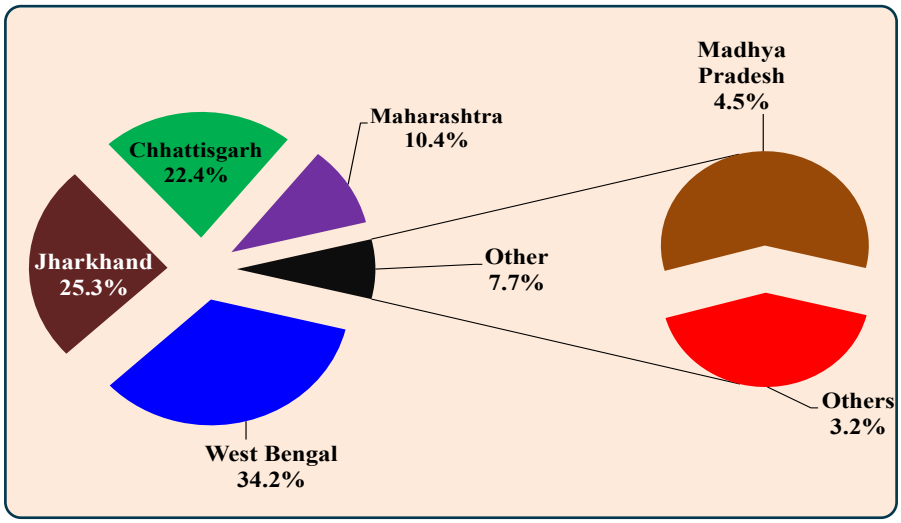


Figure 11. States-wise share in lac processing during 2017-18

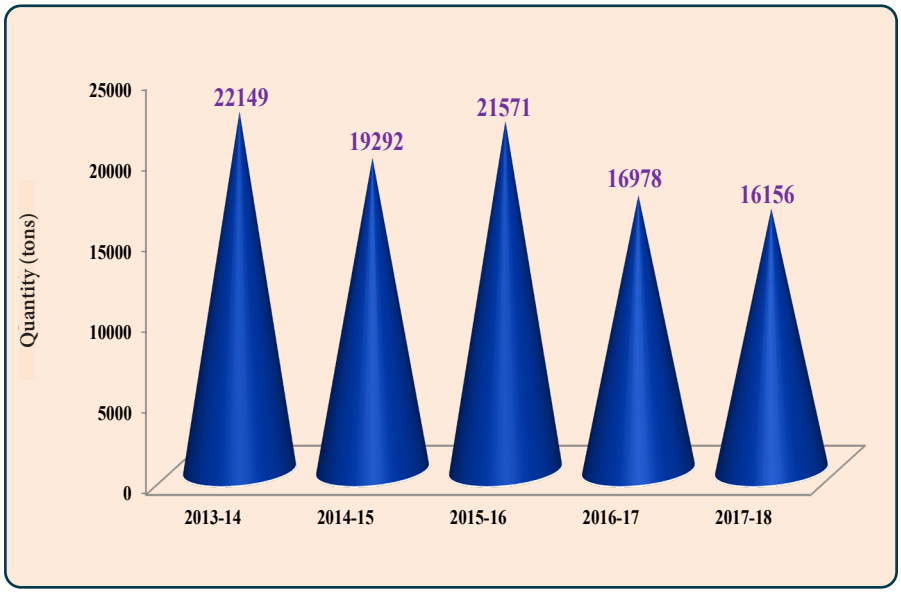


Figure 12. Lac processing in India during previous years

Table 7. Major lac processing centers in India during 2017-18

States	Districts / Centres	No. of processing units	Products made
Chhattisgarh	Dhamtari	15	Seedlac, Button lac, Bleached lac, Aleuritic acid
	Janjgir-Champa	5	Seedlac, Shellac, Bleached lac, Dewaxed Shellac, Lac dye
	Kanker	2	Seedlac
	Korba	7	Seedlac, Shellac, Bleached lac, Button lac
	Rajnandgaon	1	Seedlac, Shellac
	Raipur	1	Bleached lac, Aleuritic acid
Jharkhand	Daltonganj	2	Seedlac
	Ranchi and Khunti	10	Seedlac, Button lac, Shellac, Lac dye, Bleached lac
	Simdega	2	Seedlac
	Saraikela-Kharsawan	1	Bleached lac
	West Singhbhum	1	Black Shellac
Madhya Pradesh	Indore	1	Seedlac, Bleached lac, Varnish
	Balaghat	2	Seedlac
	Seoni	2	Seedlac
	Hosangabad	1	Seedlac
Maharashtra	Gondia	6	Seedlac, Shellac, Gasket Shellac Compound, Bleached lac
West Bengal	Purulia	84	Seedlac, Shellac, Button lac, Bleached lac, Aleuritic acid, lac wax, Dewaxed decolourised lac
	North 24 Paragana	1	Aleuritic acid
	Others	20	Lac based value added products
Others		10	Lac based value added products
TOTAL		**174**	

Table 8. Quantity of sticklac processed in India during 2017-18

States	Districts/ Centres	Quantity processed (tons)	% change over last year
Chhattisgarh	Dhamtari	1110	-21.3
	Janjgir-Champa	1940	12.8
	Kanker	9	-96.1
	Korba	310	-82.2
	Rajnandgaon	250	38.9
	Sub total	**3619**	**-31.3**
Jharkhand	Daltonganj	75	66.7
	Ranchi	315	55.2
	Khunti	3630	-12.0
	Saraikela-Kharsawan	0	-100.0
	Simdega	71	-29.2
	West Singhbhum	0	-100.0
	Sub total	**4091**	**-11.0**
Maharashtra	Gondia	**1685**	**-5.1**
Madhya Pradesh	Indore	100	11.1
	Balaghat	250	257.1
	Seoni	300	100.0
	Hoshangabad	70	0.0
	Sub total	**720**	**89.5**
West Bengal	Purulia	4669	28.6
	North 24 Paragana	350	0.0
	Others	505	-30.8
	Sub total	**5524**	**17.3**
Others	Delhi, Gujarat, Odisha, Karnataka, Tamil Nadu, Uttar Pradesh, etc.	518	108.8
	Total	**16156**	**-4.8**

Table 9. Markets with annual arrival of over 500 tons

States	Primary markets	Secondary markets
Chhattisgarh	Bhaisama Bazar	Dhamtari, Kathgora and Sakti
Jharkhand	Tapkara, Bandgaon, Jaldega and Kolebira	Khunti
Maharashtra	Barghat region	Gondia
West Bengal	Balarampur, Jhalda and Tulin	Balarampur

Name of the primary and secondary markets with annual arrival of over 500 tons and district wise distribution of lac processing centres across major lac producing districts of India have been presented in Table 9 and Table 10, respectively.

Table 10. Lac processing centers in the major lac producing districts of India

Classifications (Qty. in tons)	No. of districts	No. of processing centers	Name of the districts
> 1,000	6	124	Khunti, (Jharkhand); Janjgir-Champa, Dhamtari and Korba (Chhattisgarh); Gondia (Maharashtra) and Purulia (West Bengal).
100-1000	6	31	Kanker and Rajnandgaon (Chhattisgarh); Ranchi and Simdega (Jharkhand); North 24 Paragana (West Bengal) and Seoni (Madhya Pradesh) and other districts (West Bengal).
1- 100	7	9	Palamu, Saraikela-Kharsawan, West Singhbhum (Jharkhand); Raipur (Chhattisgarh); Indore, Balaghat and Hoshangabad (Madhya Pradesh)
< 1	7	10	Rest of the districts in Assam, Andhra Pradesh, Bihar, Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Meghalaya, Odisha, Uttar Pradesh, Telangana and West Bengal.
Total	**26**	**174**	

Trends in price of lac over the time

Time series data on prices of *rangeeni* and *kusmi* sticklac and seedlac was collected from the above mentioned lac markets of Jharkhand, West Bengal, Chhattisgarh, Maharashtra and Madhya Pradesh on quarterly basis. The prices of *rangeeni* and *kusmi* sticklac as well as seedlac has shown increasing trend from December, 2010 to December, 2012. TRIFED launched a mega scheme to safeguard the interest of the forest dwellers by providing Minimum Support Price (MSP). Subsequently, market price also started to rise up from December 2015 from its lowest level. In comparison of this scenario, it is evident from the Figure 13 that local market price of lac improved significantly till December 2017.

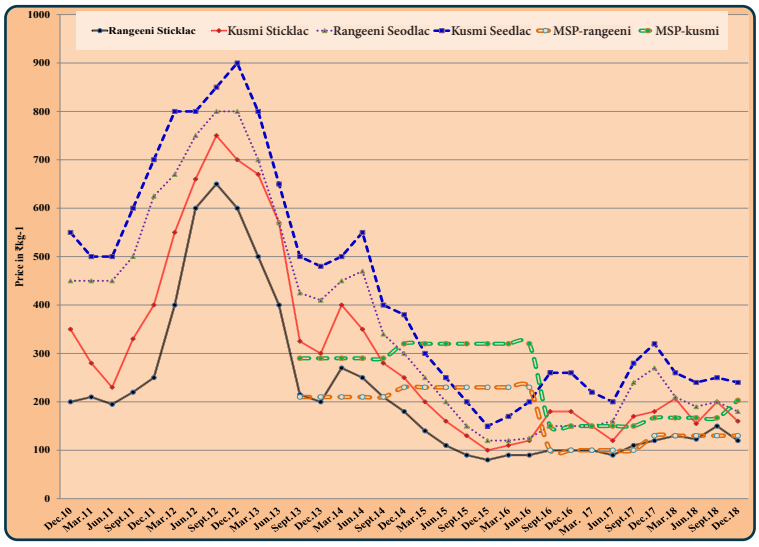


Figure 13. Trends in price of sticklac and seedlac over the time

International trade of lac and its value added products

Data on export of lac and its value added products from India were collected from Shellac and Forest Products Export Promotion Council (SHEFEXIL), Kolkata. The total export of lac and its value added products during the year 2017-18 was 6889.48 tons which was valued ₹225.90 crores. Details of export in quantity and value, list of top importing countries of Indian lac and export of lac has been presented in the Table 11 and Table 12 while direction of the trade, the trend in export of lac in quantity and value during last seven years and share of different items of lac export value from India are shown in Figure 14, Figure 15 and Figure 16, respectively.

Table 11. Export of lac and its value added products from India during 2016-18

Name of product	Export in 2016-17		Export in 2017-18	
	Quantity (tons)	Value (₹ lakh)	Quantity (tons)	Value (₹ lakh)
Shellac	3273.16	8556.63	2246.04	6620.66
Aleuritic acid	151.76	2658.47	157.23	3223.96
Seedlac	3151.45	6109.3	2925.62	6977.43
Dewaxed shellac	274.81	1268.07	185.23	987.75
Bleached lac	385.06	2068.67	291.07	1704.14
Shellac wax	4.49	28.7	10.88	90.92
Buttonlac	0.00	0.00	1069.95	2975.07
Dewaxed decolourised	0.00	0.00	0.00	0.00
Gasket lac	0.00	0.00	3.46	10.07

Name of product	Export in 2016-17		Export in 2017-18	
	Quantity (tons)	Value (₹ lakh)	Quantity (tons)	Value (₹ lakh)
Lac dye	0.00	0.00	0.00	0.00
Dewaxed bleached shellac	0.00	0.00	0.00	0.00
Garnet shellac	0.00	0.00	0.00	0.00
Dewaxed garnet shellac	0.00	0.00	0.00	0.00
Shellac(kiri)	0.00	0.00	0.00	0.00
Stick lac	0.05	0.08	0.00	0.00
Waxy bleacedlac	0.00	0.00	0.00	0.00
Total	**7240.78**	**20689.92**	**6889.48**	**22590.00**

Table 12. Top importing countries of Indian lac during 2017-18

Country	Quantity (tons)	Value (US $ Millions)	% Share
Bangladesh Pr	1474.25	6.21	18.2
Pakistan Ir	1560.58	5.98	17.5
U S A	876.72	3.85	11.3
Germany	783.55	3.70	10.8
Switzerland	92.00	2.68	7.8
China P Rp	233.95	2.38	7.0
Iraq	378.00	1.57	4.6
Indonesia	254.35	1.11	3.3
Egypt A Rp	242.75	0.94	2.8
Canada	192.65	0.80	2.3
Spain	145.63	0.75	2.2
U K	93.94	0.68	2.0
Italy	139.00	0.62	1.8
Netherland	10.30	0.36	1.1
Australia	37.30	0.31	0.9
Japan	62.20	0.29	0.9
France	37.20	0.29	0.8
Bolivia	47.70	0.22	0.7
Haiti	46.25	0.20	0.6
Others (32)	181.17	1.23	3.4
Total	**6889.48**	**34.15**	**100.00**

Among the export destinations of lac and its value added products Bangladesh (18.2%), Pakistan (17.5%), USA (11.3%), Germany (10.8%), Switzerland (7.8%) and China (7.0%) remained top destinations with more than 70% import of lac and its value added products from India. Iraq (4.6%), Indonesia (3.3%), Egypt (2.8%), Canada (2.3%), Spain (2.2%), UK (2.0%), Italy (1.8%) and Netherland (1.1%). At global level, Indian lac was exported to 51 countries of Europe, America, Asia, Africa and Australia continents.

Lac is an important natural resins exported by and also imported from Indonesia, Thailand, *etc.* in India. During 2017-18, exported value showed that 99.37 % contribution in total export of natural resins were from lac (91.72 %), gum rosin (3.84 %) and other resins (3.81 %).

Table 13. Export of lac during XI and XII Plans periods

Year	Quantity (tons)	Value (₹ lakh)
XI Plan		
2007-08	7906.33	12426.87
2008-09	6968.42	12414.50
2009-10	6422.61	11002.33
2010-11	6339.05	21112.92
2011-12	6858.21	36461.30
Average	**6898.92**	**18683.58**
XII Plan		
2012-13	4361.30	48027.58
2013-14	8158.10	56853.63
2014-15	6569.17	32249.58
2015-16	7668.42	24755.18
2016-17	7241.00	20690.00
Average	**6799.60**	**36515.19**
2017-18	6889.48	22590.00

Average annual export quantity of lac during last ten (2010-11 to 2017-18) years was 6,760.59 tons valued ₹32842.52 lakh. Average annual demand of lac was 6,799.60 tons valued ₹36515.19 lakh during the period 2012-13 to 2016-17. However, during last eight years (2010-11 to 2017-18) Annual compound growth rate in exported quantity was positive for lac (2.81%) and total natural resins group (1.67%) as the acceleration in quantity demanded since 2012-13 onwards. But, in value terms the same figure has shown decelerating growth rates.

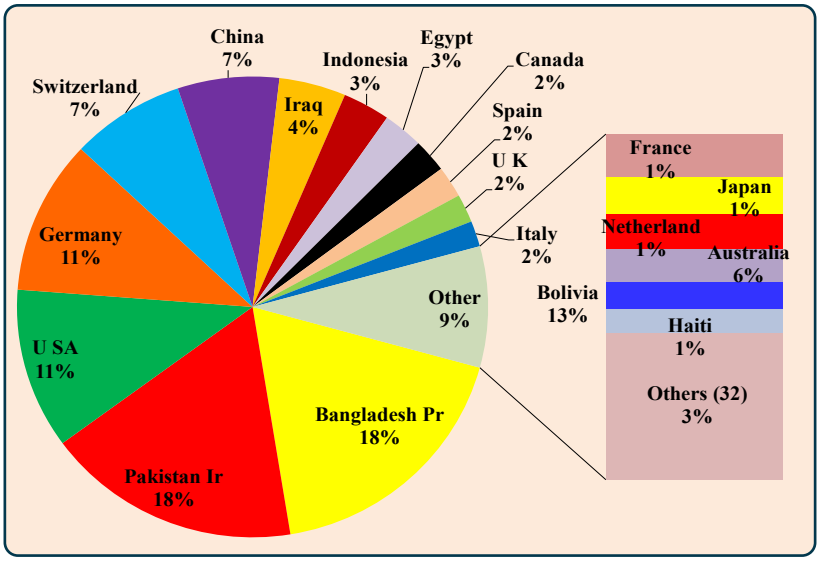


Figure 14. Direction of the trade of lac and its value added products

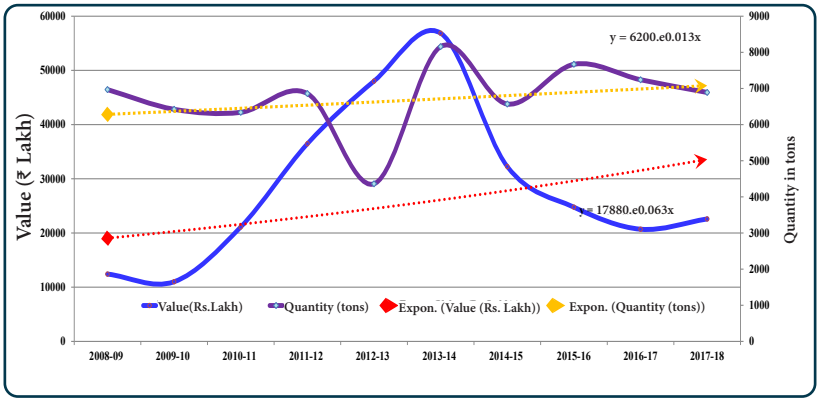


Figure 15. Trends in export of lac based products from India

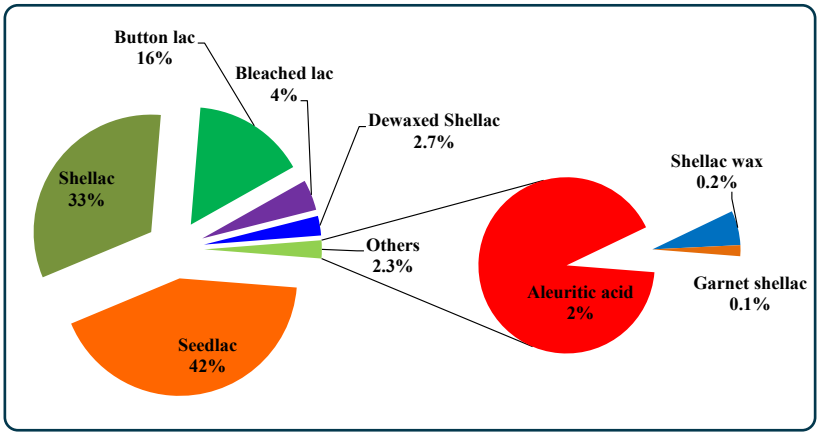


Figure 16. Product-wise export share of various lac based products

Pine resin –a flora based resin

Aggregate World trade of rosin and resin acids is about 1100 million US dollars. Zambia, Indonesia, USA, Brazil, Austria, Finland, Russian Federation and Sweden are the major suppliers. Pine resins are secretion of pine resins (*Pinus roxburghii* Sarg.), particularly coniferous trees. These are used in the production of varnishes, adhesives, and food glazing agents. *Chir* pine yields commercially important oleo-resin which forms the raw material for rosin and turpentine oil industry in India. *Chir* pine is widely tapped for resin on commercial basis, particularly in the hills of Himachal Pradesh, Uttarakhand, Jammu & Kashmir and north-eastern states. The northern hill states annually produce around 8,000 to 9,000 tons of raw resins extracted from pine trees.



Photo 2. Natural resin: Plant Kingdom

Commercially tapped sources of pine resin are depicted in Table 14.

Table 14. Commercially tapped sources of pine resin: species and country of production

Sl. No.	Major species	Producing country
1.	*Pinus caribaea* Morelet	Venezuela, South Africa, Kenya
2.	*P. halepensis* Miller	Greece
3.	*P. kesiya* Royale ex Gordon	China
4.	*P. massoniana* D. Don	China
5.	*P. merkusii* Jungh. & Vriese	Indonesia, Vietnam
6.	*P. oocarpa* Schiede	Mexico, Honduras
7.	*P. pinaster* Aiton	Portugal
8.	*P. radiata* D. Don	Kenya
9.	*P. roxburghii* Sarg.	India, Pakistan
10.	*P. sylvestris* L.	Russia
11.	*P. elliottii* Engelm.	Brazil, Argentina, South Africa, USA, Kenya

Source: *FAO, Rome, Italy*

In sub-tropical India, pine forests are found in Bhutan, North India, Kashmir, Nepal, Pakistan, Sikkim and southern part of Tibet with altitudes ranging between 400 and 2500 m (Naithani, 1984; Tewari, 1994). The hills of Uttarakhand, in the northern India are one of the best habitats of chir pine forests. Five species of pines are indigenous to India viz. *P. roxburghii* (Chir pine), *P. wallichiana* (Blue pine), *P. kesiya* (Khasi pine), *P. gerardiana* (Chilgoza pine) and *P. merkussi* (Teriasserian pine) (Gamble, 1902). Out of these *P. roxburghii*, *P. wallichiana* and *P. gerardiana* are found in the Himalayas, whereas *P. kesiya* and *P. merkussi* are indigenous to Assam (India), China and Burma. Oleoresins from pines are composed of two components,

volatile turpentine oil and the remaining solid transparent material called as Rosin. The greatest single use of turpentine in India is for the production of synthetic camphor.

Pine resin production

In Asia, China and Indonesia were leading producers of pine resin followed by India. Total worldwide production of rosin, which is made up of resin extracted from pine, was about 1.2 million tons per annum in 1995 (FAO). Global trade of pine resins ranged from 3-4 lakh tons only in 2017. As per FAO reports (Coppen and Hone, 1995), crude resin production in India peaked at about 75000 tons in 1975-76 and has since fallen steadily. Production in 1990-91 was less than 25000 tons. The main reason for the decline was the loss of trees for tapping, either because many of them had reached the end of their productive lives and there were no new areas of pine with which to replace them, or because the damage done to trees by the use of inefficient, incorrectly applied methods of tapping led to Forest Department bans on tapping.

Major share of resin production comes from Himachal Pradesh and Uttarakhand. The production of resin in the states during the year 2017-18 was about 5000 tons and more than 85 % of this raw material is processed in the Rosin and Turpentine oil Factories (RTFs). Indonesia also supplies rosin to Indian industries but China is the major supplier. Indonesian rosin costs around the same as the Chinese product. Both China and Indonesia have captured more than 50% of the Indian market. After the global recession, China had dropped its prices in the Indian market. The current demand of rosin in the country is ranged from 40,000 to 50,000 tons annually. In India, resin yield varied between 4 and 7 kg per tree. In India resin production and its value during 2010-11 to 2017-18 is depicted in Figure 17.

Resin production policy

In Uttarakhand (earlier known as Uttar Pradesh) the tapping of resin began in 1890 that was commercialized in 1896. It was extended to Jammu and Kashmir and Himachal Pradesh in 1940 and 1945, respectively (Singh and Asokan, 1984).

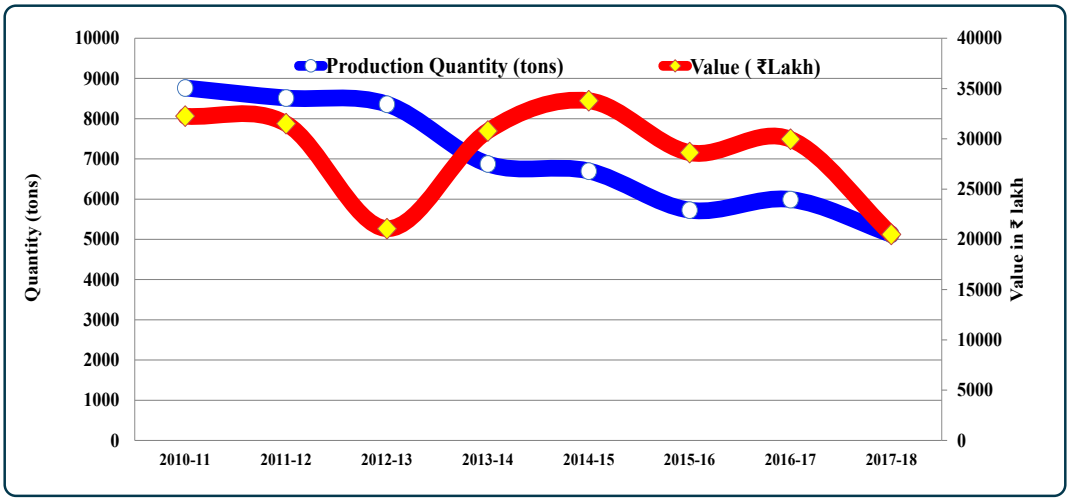


Figure 17. Pine resin production and its value in Himachal Pradesh and Uttarakhand

The old method used for resin tapping was the faulty French "cup and lip" method, which was replaced by the "rill" method (Kaushal and Khosla, 1984; Chaudhari *et al.*, 1988). For the Himalayan region, extraction of resin as a source of revenue is reported since year 1911 (Nautiyal *et al.*, 2006). State wise annual pine resin production and their contribution in the total production is given in Table 15 and graphical illustration presented in Figure 18.

Table 15. State wise annual pine resin production and their contribution in the total production

Year	Uttarakhand	Himachal Pradesh	Total
2010-11	1764 (20.1)	7000 (79.9)	8764 (100.0)
2011-12	1614 (19.0)	6900 (81.0)	8514(100.0)
2012-13	1467 (17.5)	6894 (82.5)	8361 (100.0)
2013-14	1486 (21.6)	5389 (78.4)	6875 (100.0)
2014-15	1442 (21.5)	5258 (78.5)	6699 (100.0)
2015-16	1562(27.3)	4164(72.7)	5726(100.0)
2016-17	1718 (28.7)	4273 (71.3)	5773 (100.0)
2017-18	1500 (28.5)	3754 (71.5)	5254(100.0)
Average	**1569 (22.3)**	**5454 (77.7)**	**7023(100.0)**

Figures in parentheses are the percentage of the total

Uttarakhand: To ensure transparency in allotment of resin to different agencies, resin policy has been formulated and issued in 2003.

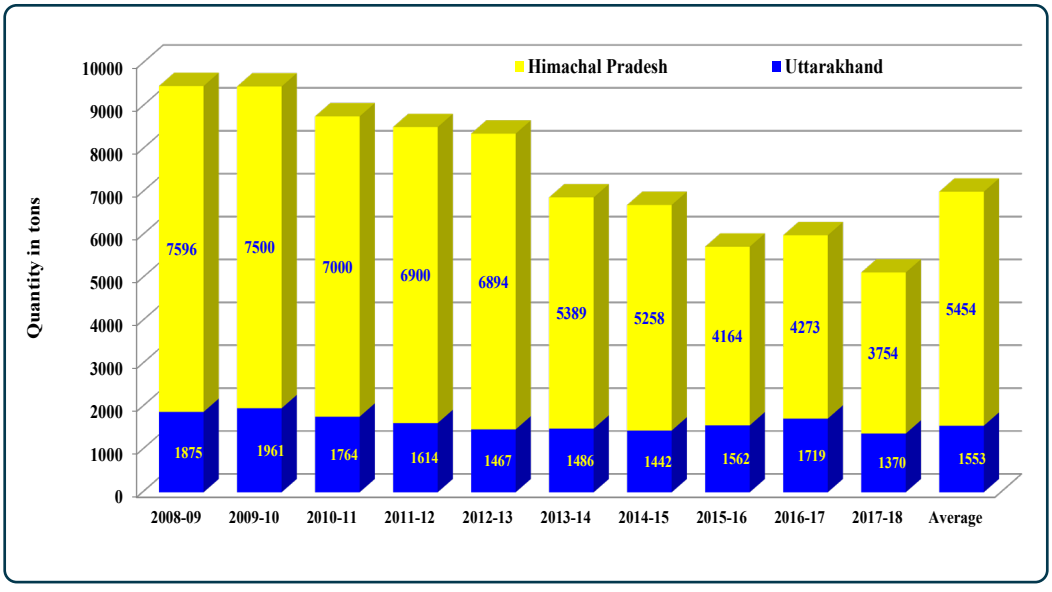


Figure 18. Pine resin production in Himachal Pradesh and Uttarakhand

Himachal Pradesh: The hill state annually produces around 6350 tons of raw rosin extracted from pine trees, grown mainly in lower hills of Hamirpur, Una, Mandi, Solan and Sirmaur districts. Exploitation of resin in the state is done by the Himachal Pradesh State Forest Corporation, Shimla. About 15.76 lakh blazes were handed over to H.P. State Forest Corporation for tapping during the year from state forests. The quantity extracted by the Corporation was 37544 quintals with the total royalty of more than ₹10.00 Crores payable by the H.P. State Forest Corporation. As per decision of the Pricing Committee of Government of Himachal Pradesh, a rate of Rs. 51/- per blaze was fixed for the year 2017(CAG Report, 2018).

Pine resin processing in India

The HP State Forest Corporation: The Himachal Pradesh State Forest Corporation functioned principally as a harvesting agency during the year. The entire exploitation works of timber, resin, fuel-wood, charcoal, *etc.* from the state forests were handled by the Forest Corporation. There are resin processing units functioning in private sector in Hamirpur Circle particularly in Una and Dehra Divisions and two in public sector at Bilaspur and Nahan.

The entire government resin tapping work is being done by the Corporation, through modern techniques (Rill method as of now). It is graded and sold on the basis of colour, the palest shades of yellow-brown being the better quality. Quality criteria and specifications are described in Table 16.

Table 16. Quality criteria for different grades of the rosin

Sl. No.	Rosin	Grade	Full Name
	Pale	X	Extra White
		WW	Water White
		WG	Window Glass
		N	Nancy
	Medium	M	Mary
		K	Kitty
		H	Harry
	Dark	D	Dark
		B	Black

The two Resin & Turpentine Oil Factories are located at Bilaspur and Nahan operational since four decades, have both skilled and unskilled workers to undertake resin processing and other diversification activities. Rosin and turpentine oil and other subsidiary products like phenyl, varnish, black japan, *etc.* are produced from the collected resin. Rosin is the main product obtained from pine resin. It remains behind as the residue after distillation of the turpentine oil. It is a brittle, transparent, glassy solid. It is insoluble in water but soluble in many organic solvents.

Pine resin market and price

Rosin is purchased on the basis of its color, which varies from pale amber to black. The lighter the colour of rosin, the more valuable it is.

Table 17. Grade-wise price of rosin during 2017-18

Rosin Category	Grade	Full name	Rate (₹/Kg)			
			Nahan-TP barrels	Nahan-PGI barrels	Bilaspur-TP barrels	Bilaspur-PGI barrels
Pale	X	Extra White	82.5	84.0	81.5	83.0
	WW	Water White	81.5	83.0	80.5	82.0
	WG	Window Glass	78.5	80.0	77.5	79.0
	N	Nancy	74.5	76.0	73.5	75.0
Medium	M	Mary	72.5	74.0	71.5	73.0
	K	Kitty	71.5	73.0	70.5	72.0
	H	Harry	70.5	72.0	69.5	71.0
Dark	D	Dark	67.5	69.0	66.5	68.0
	B	Black	57.5	59.0	56.5	58.0

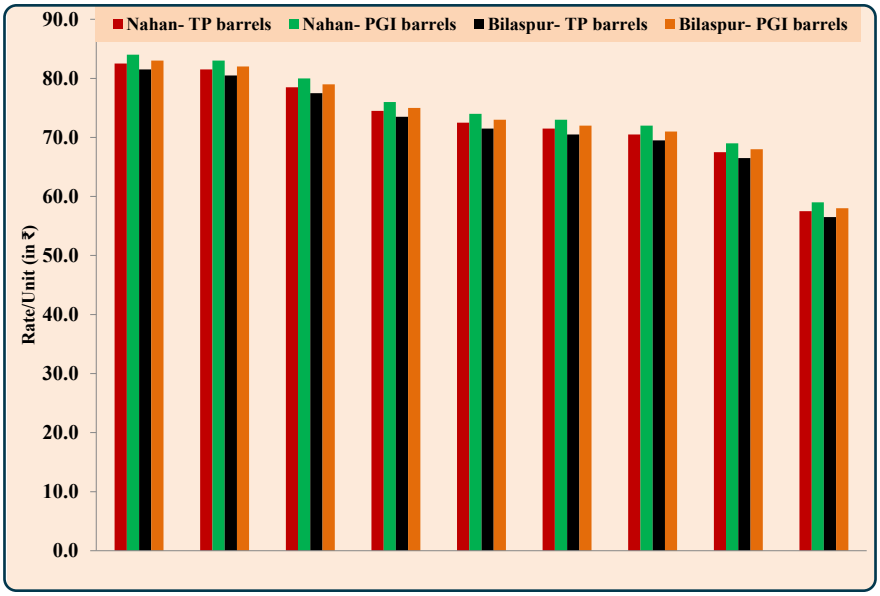


Figure 19. Price pattern of various grades of Pine resin based products

Price of turpentine oil (vegetable) per litre (naked) ex-factory on cash basis ranged from ₹ 56.5 to ₹ 84.0. The price of per kilogram rosin is given in Table 17 and graphically shown in Figure 19. This price is an ex-factory price including packing charges excluding taxes and duties *etc.*

International trade of pine resin

Details of export in quantity and value, list of countries importing resin from India and details of resin supply from various countries during 2017-18 is presented in the Table 18.

Table 18. Destination and source wise trade scenario of rosin during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
A. Export				
Iran	134.40	118.66	180.00	39.13
Pakistan Ir	40.80	35.68	60.00	13.04
Cote d' Ivoire	30.00	45.57	70.00	15.22
Lebanon	25.00	38.29	60.00	13.04
U Arab Emts	22.40	22.76	40.00	8.70
Sri Lanka Dsr	9.12	8.73	10.96	2.38
Saudi Arab	8.00	6.88	10.00	2.17
U S A	3.09	5.58	9.00	1.96
Yemen Republic	3.00	3.77	9.00	1.96
Bhutan	6.27	2.50	3.00	0.65
Nepal	3.00	2.10	3.00	0.65
Israel	1.92	1.71	2.00	0.43
Singapore	1.00	1.41	2.00	0.43
Others (3)	0.42	0.61	1.04	0.23
Total	**288.42**	**294.25**	**460.00**	**100.00**
B. Import				
Indonesia	14,929.15	12,780.55	19,830.00	31.29
Brazil	13,486.00	11,774.51	18,280.00	28.84
Nepal	13,224.60	9,952.92	15,450.00	24.38
China P Rp	3,132.20	4,428.02	6,880.00	10.86
Vietnam Soc Rep	976.90	733.22	1,140.00	1.80
Argentina	366.40	284.67	440.00	0.69
Mexico	354.83	258.24	400.00	0.63
Japan	96.03	216.89	340.00	0.54
U S A	55.68	209.81	330.00	0.52
Belgium	35.50	102.60	160.00	0.25
France	7.00	25.78	40.00	0.06
Germany	19.49	27.28	40.00	0.06
Portugal	10.00	19.27	30.00	0.05
Others (4)	9.85	22.10	20.00	0.03
Total	**46,703.63**	**40,835.86**	**63,380.00**	**100.00**

The total export of rosin during the year 2017-18 was 288.42 tons which was valued ₹294.25 lakh.

Similarly, total import of rosin during the year 2017-18 was 46703.63 tons which was valued ₹40835.86 lakh.

The analysis of EXIM data revealed that the exported quantity of rosin increased to 102.2% while imported quantity of rosin was increased by 8.37% in 2017-18 in comparison to 2016-17. During this year in international market, export price of rosin decreased by 19.02 %, and import price decreased by 4.82 % comparatively of previous year. Demand of resin was higher than the quantity processed in India. Hence, India depends on Indonesia, Brazil, Nepal, China and Vietnam to meet its domestic demand. About 99 % of gum rosin was supplied from these five neighboring countries. About 98 % gum rosin is exported to Iran, Pakistan, Cote D' Ivoire, Lebnan, U Arab Emts, Sri Lanka, Saudi Arab, U S A and Yemen Republic. The destination and source wise demand and supply of the rosin is depicted in the Figure 20 and Figure 21. Rosin is an important natural resin in the basket of import items.

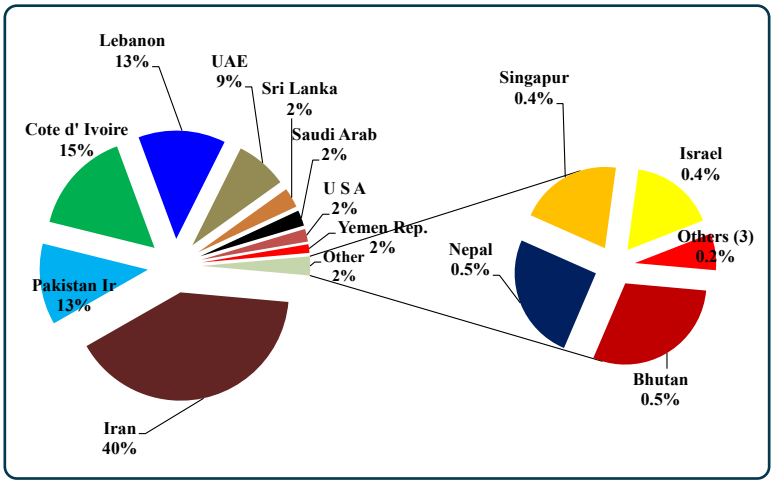


Figure 20. Destination wise scenario of rosin during 2017-18

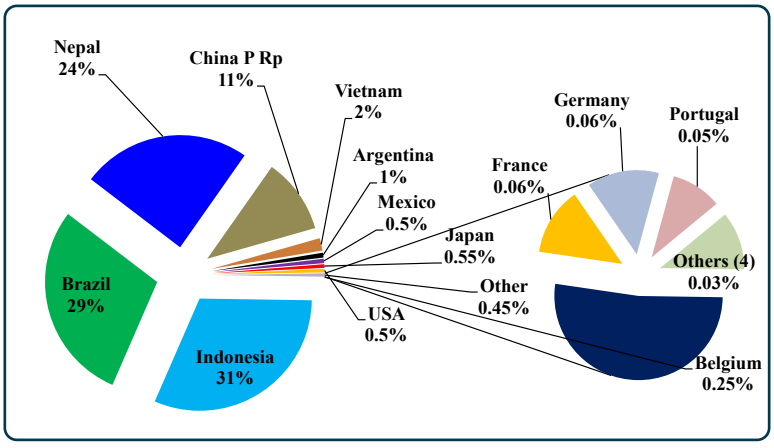


Figure 21. Source wise scenario of rosin during 2017-18

Copal

Copals are derived from species of *Bursera*, *Protium* (*Burseraceae*) and *Hymenaea* and have adhesive property. It is evident from Table 19 that copal was supplied in India from Indonesia (99.37%), Philippines (0.60%) and Italy (0.03%) during 2017-18. A very little portion (0.6%) of the total imported quantity was exported to only eleven countries during 2017-18.

Table 19. Destination and source wise trade scenario of copal during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000' US dollars)	Share (%)
Export				
Pakistan IR	6.00	2.73	4.18	37.92
Nepal	3.50	1.81	2.77	25.14
Canada	1.35	0.92	1.41	12.78
Mexico	0.75	0.76	1.16	10.56
U S A	0.09	0.31	0.47	4.31
U K	0.22	0.25	0.38	3.47
Uruguay	0.07	0.23	0.35	3.19
Belgium	0.02	0.08	0.12	1.11
Germany	0.14	0.08	0.12	1.11
Australia	0.02	0.02	0.03	0.28
U Arab EMTS	0.01	0.01	0.02	0.13
Total	**12.17**	**7.20**	**11.03**	**100.00**
Import				
Indonesia	2,000.85	1,612.13	2500.00	99.37
Philippines	27.75	9.79	15.00	0.60
Italy	0.01	0.48	0.70	0.03
Total	**2,028.61**	**1,622.40**	**2515.70**	**100.00**

Dammar batu

Damar is tapped from the *sal* tree *(Shorea robusta).* Annual production of Dammar batu in India is about 80-100 tons and used as painting and incense material. Destination wise EXIM scenario of dammar batu during 2017-18 is given in Table 20. During 2017-18, more than 95% of this resin in India was supplied from Indonesia (58.25%) and Thailand (37.11%). A very little portion (0.2%) of the total imported quantity was exported to mainly Indonesia (46.96%), Israel (20.85%), Jordan (13.65%) and Malaysia (11.85%).

Table 20. Destination wise trade scenario of dammar batu during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000' US dollars)	Share (%)
Export				
Indonesia	27.00	15.65	23.36	46.96
Israel	3.00	6.95	10.37	20.85
Jordan	2.69	4.55	6.79	13.65
Malaysia	1.55	3.95	5.90	11.85
Australia	0.10	0.96	1.43	2.88
Germany	0.25	0.85	1.27	2.55
Kenya	0.25	0.16	0.24	0.48
Bangladesh Pr	0.50	0.15	0.22	0.45
Bhutan	0.10	0.08	0.12	0.24
Italy	0.01	0.03	0.04	0.09
Total	**35.45**	**33.33**	**49.75**	**100.00**
Import				
Indonesia	15251.88	5326.87	7950.55	58.25
Thailand	4181.79	3393.68	5065.19	37.11
Vietnam Soc Rep	655.59	310.47	463.39	3.40
Lao Pd Rp	105.00	104.02	155.25	1.14
Malaysia	33.00	9.39	14.01	0.10
Total	**20227.26**	**9144.43**	**13648.40**	**100.00**

Mastic gum

This is the dried exudate of the shrub-like tree *Pistacia lentiscus L. var. chia* of the *Anacardiaceae* family, cultivated exclusively in the south of the Greek island Chios. It has the application in dentistry, photographic negatives and paints industry. In India, it is imported from Greece (Table 21) and there was no export of mastic gum during 2017-18.

Table 21. Destination wise trade scenario of mastic gum during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000' US dollars)	Share (%)
Import				
Greece	2.2	11.59	20.00	100.00
Total	**2.2**	**11.59**	**20.00**	**100.00**

B. NATURAL GUMS

Plant exudate based natural gums



Photo 3. Natural gums: Gum processing

Procurement data available with Odisha Forest Development Corporation Ltd, Rajasthan Tribal Areas Development Coop Federation Ltd, Maharashtra State Cooperative Tribal Development Corporation Ltd, Jharkhand State Minor Forest Produce Cooperative Development Marketing Federation Ltd, Girijan Co-operative Corporation Limited, Visakhapatnam, Andhra Pradesh, Chhattisgarh M.F.P. (T&D) Fed. Ltd., Raipur, Chhattisgarh, Gujarat State Forest Development Corporation Limited (GSFDCL), Vadodara was recorded and compiled. Gum tapping is mainly done in the schedule areas where tribal populations exist. The collection charges to the collectors at collection centres are paid by the purchaser at the rate fixed by the Govt. After making the payment of the collection charges and the difference amount of sale rate and collection rate in the District Union, the purchasers are allowed to transport the collected gums wherever they desire. In India, mainly *karaya* gum (*S. urens*), *dhawada* gum (*A. latifolia*), *prosopis* gum (*P. juliflora*), *khair* gum (*A. catechu*), *babool/babul* gum (*A. nilotica*), *Jhingan* (*Lannea coromandelica*), *palas* (*B. monosperma*), *char* (*B. lanzan*) and *guggul* gum (*C. wightii*) are produced.

About 80 % of natural gum production in the country is contributed by Maharashtra (37.3 %), Madhya Pradesh (18.4 %), Jharkhand (11.1 %), Telengana (6.6 %), and Chhattisgarh (5.7 %). Rest of the 20 % comes from Gujarat (4.2 %), Andhra Pradesh (3.4%) and other minor gum producing states (13.2%). District wise gum producing areas in the country are presented in Annexure III. In parts of the Jaipur, Ajmer and Jodhpur districts *A. senegal* is common. *A. catechu* forests are common in the south-eastern regions. *e.g.* Baran, Jhalawar, Kota, Tonk , Chittorgarh and Alwar. The state wise gum production during 2012-13 to 2017-18 is shown in Table 22 and graphical illustration is shown in Figure 22.

Table 22. Gum production in major gum producing states of India (in tons)

Year	Maha-rashtra	Madhya Pradesh	Jhark hand	Telan-gana	Chhat-tisgarh	Gu-jarat	Andhra Pradesh	Oth-ers	Total
2012-13	350.8	232.4	207.3	102.2	23.6	42.6	29.9	155.2	1144.0
2013-14	539.1	207.5	90.0	65.0	2.5	54.7	42.5	143.0	1144.4
2014-15	323.8	120.0	56.0	42.5	4.0	51.5	26.2	87.0	710.9
2015-16	221.0	113.0	56.3	33.6	139.0	16.1	19.8	85.0	683.8
2016-17	158.0	113.0	64.0	40.0	73.0	16.1	10.29	95.0	569.3
Average	**318.5**	**157.2**	**94.7**	**56.7**	**48.4**	**36.2**	**25.7**	**113.0**	**850.5**
2017-18*	66.0	30.0	19.0	12.0	15.0	6.1	5.0	35.0	188.1

*Note: These are the revised data set and may differ with previous estimates due to validation and updates;*Figures are based on the available information with state level federations.*

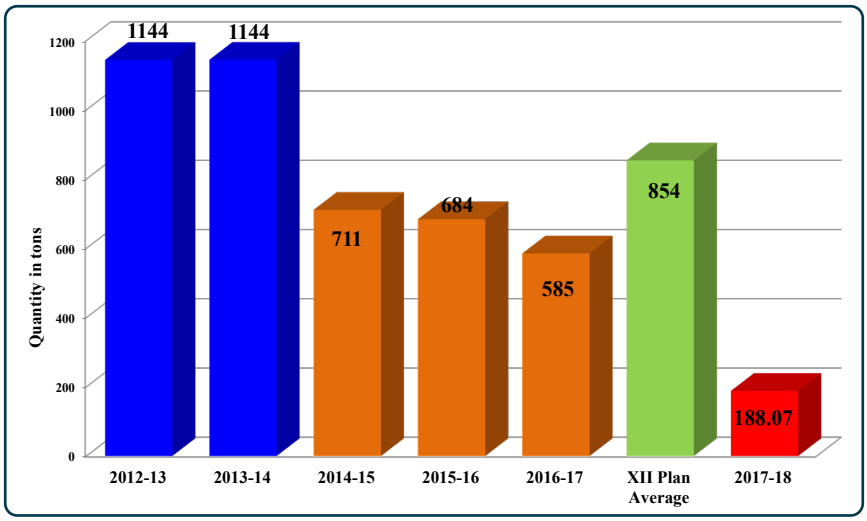


Figure 22. Quinquennial comparison of the production of natural gums in India

Gum *karaya* production in India

Gum *karaya* is the dry exudate of *S. urens* and *S. villosa*. It is also known by the name Indian tragacanth, as it resembles gum tragacanth produced by *Astragalus spp*. Gum *karaya* is one of the least soluble gums used for many industries including pharmaceutical, food, paper, textiles, cosmetic industry, *etc.*

Overall production has decreased from 211.9 tons in 2012–13 to 31.6 tons in 2017-18. During this period the price increased from ₹7.4/kg to ₹110/kg. The gum production across various states has been presented in Table 23 and Figure 23.

Table 23. *Karaya* gum production in major gum producing states of India (in tons)

State / Year	Telangana	Chhattisgarh	Andhra Pradesh	Madhya Pradesh	Jharkhand	Maharashtra	Gujarat	Others	Total
2012-13	101.6	19.1	16.3	8.2	51.5	10.0	0.2	5.0	211.9
2013-14	64.4	1.8	27.9	12.1	4.4	13.0	0.1	5.0	128.8
2014-15	39.0	4.0	16.0	10.0	4.0	5.0	0.2	5.0	83.2
2015-16	30.1	41.0	9.9	5.0	2.3	7.0	0.1	5.0	100.4
2016-17	35.0	21.0	4.1	8.0	2.0	8.0	0.1	5.0	83.2
Average	**54.0**	**17.4**	**14.9**	**8.7**	**12.8**	**8.6**	**0.1**	**5.0**	**121.5**
2017-18	10.0	3.0	0.6	5.0	2.0	6.0	0.1	5.0	31.6

The gum producing forest divisions in the Chhattisgarh are Bilaspur (Mugeli, Dindori, Ratanpur, Takhatpur, Lormi), Raipur, E. Surguja (Balarampur), Marvahi (Kota), S. Surguja, Raigarh (Khamariya,), Dharmajaygarh, Rajnandgaon, Mahasamund, Dhamtari, Korea, Sukma, Bijapur, Dantewada and W. Bhanupratapur. In Jharkhand, the *karaya* gum is produced

in the Latehar (Garu, Mahuadar, Herhanj, Balumath, Barwadih, Lesliganj, Chhipadohar and Richughutu), Chatra (Lawalang, Pratappur and Kanti), Garhwa (Ramkanda and Bhandaria), Daltonganj (Panki and Chhatarpur) and West Singhbhum (Chakradharpur). A trend in annual production of gum *karaya* in India is illustrated in Figure 24.

Collection and grading of gum *karaya*

During 2017-18, a Minimum Support Price (MSP) of ₹11600/qt for gum *karaya* was declared by Pricing Cell, TRIFED, Ministry of tribal affairs, Govt. of India.

However, prevailing market price of gum *karaya* is ranged from ₹350/kg to ₹450/kg. The scheme is initially being implemented in the states having scheduled areas and scheduled tribes in the fifth schedule of the Constitution of India (except Himachal Pradesh). Now, it is extended over the country.

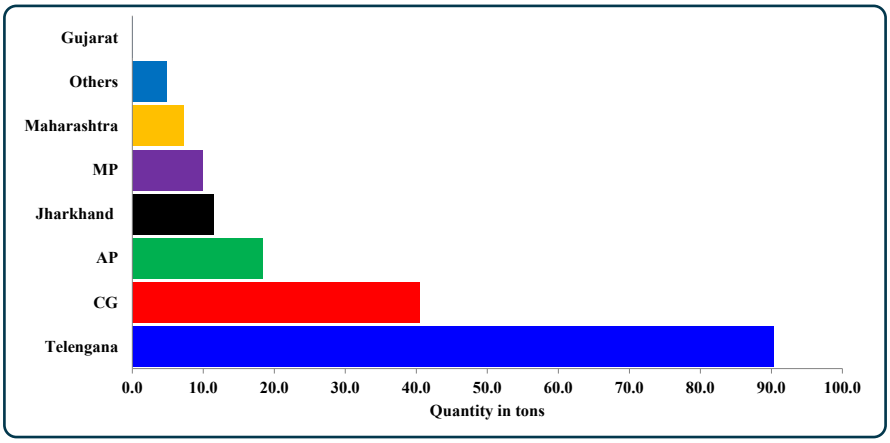


Figure 23. State-wise average production of gum *karaya* (2008-09 to 2017-18)

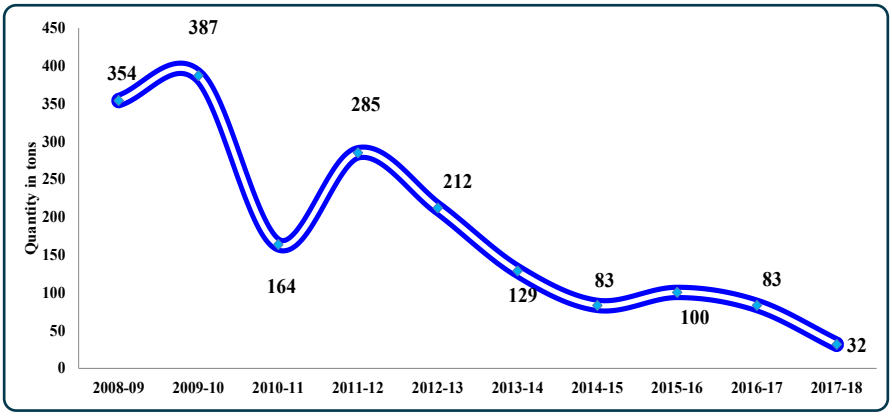


Figure 24. Trends in production of gum *karaya* in India

International trade of gum *karya*

Export of gum *karaya* from India during previous years have been depicted in Figure 25 while domestic demand from and overseas supply in India have been given in Table 24. Japan (30.66%), France (21.64%), Pakistan (8.17%), Vietnam (5.10%), Spain (4.76%), Malaysia

(4.22%), USA (3.71%), Myanmar (3.55%), China (3.17%), Germany (2.61%), Cuba (2.28%), UK (2.02%) and Belgium (1.87%) remained as major export destinations during 2017-18. Due to decline in production and comparatively cheaper rates in international market, a total of 2299.99 tons of gum *karaya* was supplied from Mali (56.03%), Senegal (37.93%), USA (6.03%) and Malaysia (0.01%) to India. Source wise import figures are presented in Figure 27.

Table 24. Export and import of gum *karaya* during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000' US dollars)	Share (%)
Export				
France	91.35	250.14	390.00	30.66
Japan	44.00	176.54	270.00	21.64
Pakistan Ir	50.05	66.68	100.00	8.17
Vietnam Soc Rep	26.50	41.61	60.00	5.10
Spain	10.00	38.86	60.00	4.76
Malaysia	13.15	34.41	50.00	4.22
U S A	9.89	30.24	50.00	3.71
Myanmar	9.00	28.99	40.00	3.55
China P Rp	17.50	25.86	40.00	3.17
Germany	8.00	21.26	30.00	2.61
Cuba	3.03	18.63	30.00	2.28
U K	6.41	16.48	30.00	2.02
Belgium	10.00	15.24	20.00	1.87
U Arab Emts	6.28	13.37	20.00	1.64
Thailand	1.10	8.75	10.00	1.07
Syria	3.00	8.56	10.00	1.05
Others (10)	4.15	20.14	50.00	2.48
Total	**313.41**	**815.76**	**1260.00**	**100.00**
Import				
Mali	2003.08	1642.63	2540.00	82.51
Senegal	285.92	218.08	340.00	10.95
U S A	5.99	96.56	150.00	4.85
Others	5.00	33.49	50.00	1.69
Total	**2299.99**	**1990.76**	**3080.00**	**100.00**

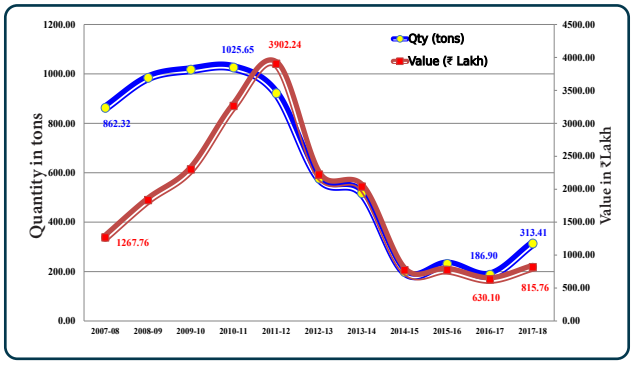


Figure 25. Trends in export of gum karaya from India

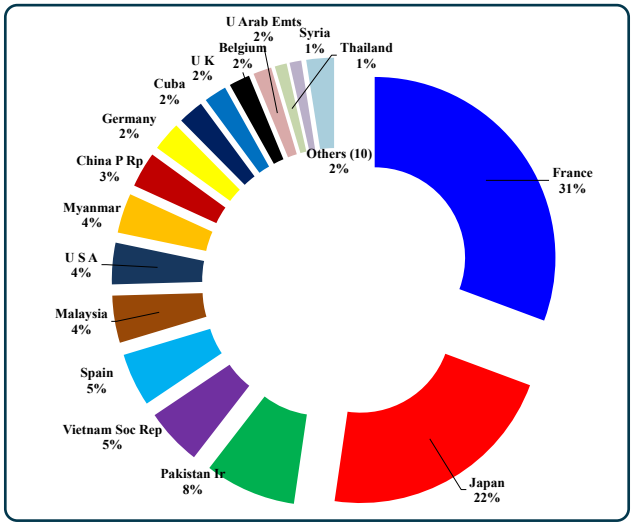


Figure 26. Destination wise scenario of gum karaya during 2017-18

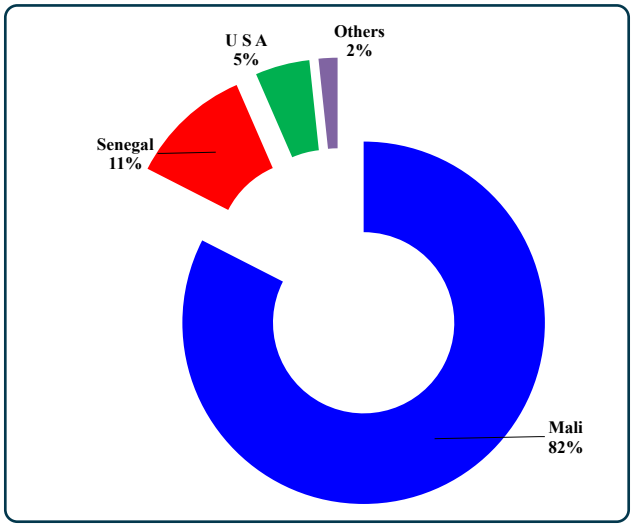


Figure 27. Source wise scenario of gum karaya during 2017-18

Dhawda gum production in India

Dhawda gum/ Gum *Ghatti* the dry exudate of *A. latifolia,* is produced in the states of Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra and Andhra Pradesh. *Dhawda* gum is used as an emulsifier of petroleum and non petroleum waxes, flavour fixative, binder in long-fibered light weight papers and stabilizer in beverages and butter containing table syrups. State-wise production data have been shown in Table 25. It is evident from the Figure 28 that total production of *dhawda* gum increased from 40.7 tons in 2007-08 to 79.00 tons in 2017-18.

Table 25. *Dhawda* gum production in major gum producing states of India (in tons)

State / Year	Andhra Pradesh*	Chhat-tisgarh	Guja-rat	Jharkhand	Madhya Pradesh	Maha-rashtra	Others	Total
2007-08	0.50	40.00	0.20	0.00	0.00	0.00	0.00	40.70
2008-09	0.60	39.30	0.20	0.00	0.00	0.00	0.00	40.10
2009-10	0.30	42.50	0.50	0.00	0.00	0.00	0.00	43.30
2010-11	0.80	1.60	0.30	0.00	0.00	0.00	0.00	2.70
2011-12	0.50	3.01	2.33	20.50	30.10	40.20	50.60	147.24
Average	**0.54**	**25.28**	**0.71**	**4.10**	**6.02**	**8.04**	**10.12**	**54.81**
2012-13	0.60	2.62	1.30	30.20	24.20	240.77	80.20	379.89
2013-14	0.80	0.20	1.40	25.40	20.80	334.11	65.70	448.41
2014-15	1.00	0.00	1.30	12.00	10.00	230.79	40.00	295.09
2015-16	1.00	6.00	1.00	4.00	8.00	134.00	40.00	194.00
2016-17	1.00	42.00	1.00	2.00	5.00	100.00	50.00	201.00
Average	**0.88**	**10.16**	**1.20**	**14.72**	**13.60**	**207.93**	**55.18**	**303.68**
2017-18	1.00	10.00	1.00	2.00	5.00	50.00	10.00	79.00

****including Telengana***

Collection and grading of *dhawda* gum

The collection rate for *dhawada* gum for the year 2017-18 was around ₹650-850/kg for Grade-I, ₹250-350/kg for Grade-II, ₹150-180/kg for Grade-III.

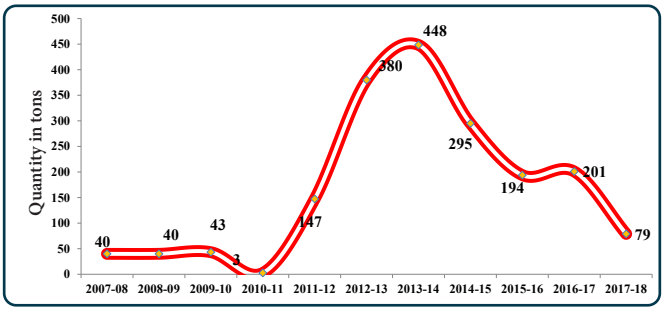


Figure 28. Year wise production scenario of gum ghatti in India

Gum *arabic* (*Acacia senegal*)

Gum arabic, also known as Acacia gum, is a natural gum made of the hardened sap of various species of the Acacia tree. Gum *arabic* is predominantly collected from two species of Acacia, namely *A. senegal* and *Vachellia (Acacia) seyal.* Producers harvest the gum commercially from wild trees, mostly in Sudan (80%). Gum *arabic* has eminent qualities as an emulsifier, stabilizer, thickener or adhesive of a non-toxic nature and its low calorific value and high soluble fibre, which are of major importance in nutrition and dietary applications. Gum production is excellent on poor soils and higher in stressed trees .Gum exudes from the duct of the inner bark and is tapped in the hot season (May-June) when the trees are under stress and annual gum production in India is about 5-10 tons.

International trade of gum *Arabic*

Sudan, Mali, Chad, Senegal and France are the major supplier to India covering more than 90% import of the gum *Arabic.*

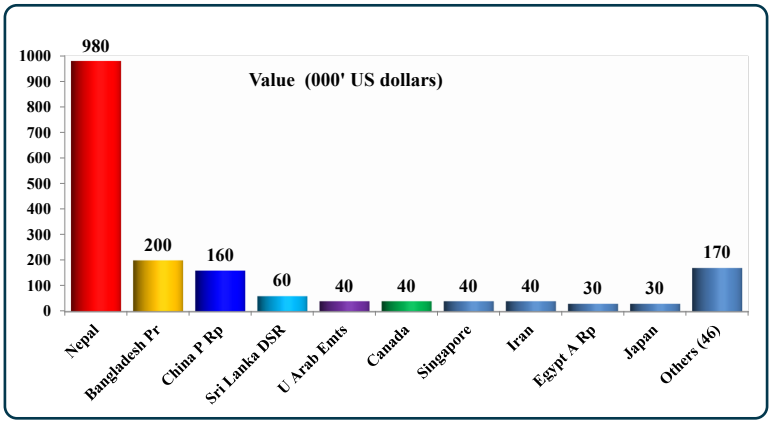


Figure 29. Destination wise scenario of gum *Arabic* during 2017-18

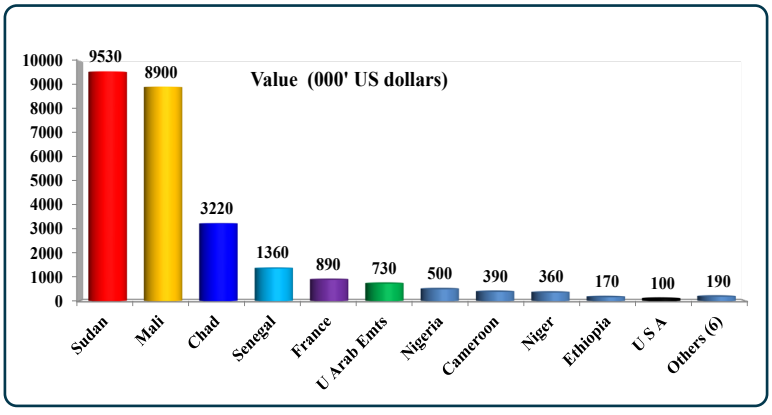


Figure 30. Source wise scenario of gum *Arabic* during 2017-18

Destination wise export and source wise import share is depicted in Table 26. There is high demand of gum *arabic* from and in India also. About 1500 tons of gum *arabic* exported to Nepal (54.71%), Bangladesh Pr (11.34%), China P Rp (8.76%), Sri Lanka (3.46%), UAE (2.42%),

Canada (2.29%) Singapore (2.04%) and Iran (2.01%) remained as major export destinations during 2017-18. About 70 % gum *arabic* supplied to these top 3 destinations (Figure 29). It was noted that Nepal emerged as major importer for gum Arabic during previous three years. During 2017-18, gum *arabic* was exported across the globe in 56 countries (Figure 30).

Table 26. Export and import of gum *Arabic* during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
Nepal	1,524.49	633.87	980.00	54.71
Bangladesh Pr	60.84	131.34	200.00	11.34
China P Rp	56.00	101.50	160.00	8.76
Sri Lanka DSR	18.05	40.07	60.00	3.46
U Arab Emts	34.28	28.08	40.00	2.42
Canada	11.32	26.59	40.00	2.29
Singapore	10.97	23.60	40.00	2.04
Iran	8.00	23.24	40.00	2.01
Egypt A Rp	4.00	19.92	30.00	1.72
Japan	5.15	17.53	30.00	1.51
Myanmar	15.87	11.97	20.00	1.03
Australia	3.84	11.47	20.00	0.99
Kuwait	6.82	8.42	10.00	0.73
Nigeria	5.57	7.78	10.00	0.67
Saudi Arab	4.95	7.48	10.00	0.65
U S A	5.41	7.23	10.00	0.62
Others (40)	38.97	58.61	90.00	5.05
Total	**1,814.53**	**1,158.70**	**1790.00**	**100.00**
Import				
Sudan	13,995.22	6,141.69	9530.00	36.19
Mali	9,860.96	5,730.71	8900.00	33.77
Chad	3,681.00	2,072.15	3220.00	12.21
Senegal	1,352.66	873.81	1360.00	5.15
France	204.51	575.60	890.00	3.39
U Arab Emts	810.56	468.12	730.00	2.76
Nigeria	857.80	319.95	500.00	1.89
Cameroon	352.00	251.02	390.00	1.48
Niger	638.00	235.49	360.00	1.39
Ethiopia	240.00	109.01	170.00	0.64
U S A	3.68	62.42	100.00	0.37
U K	14.35	52.79	80.00	0.31
Italy	19.70	51.79	80.00	0.31
Benin	44.00	14.98	20.00	0.09
Others (4)	1.07	9.31	10.00	0.05
Total	**32,075.51**	**16,968.84**	**26340.00**	**100.00**

International trade of *Asian* gum

There is high demand of *Asian* gum from and in India also. Destination wise export and source wise import share of *Asian* gum is depicted in Table 27 and graphical presentation of direction of trade is illustrated in Figure 31.

During 2017-18, about 498 tons of *Asian* gum was exported from the country. Germany (50.31 %), Russia (8.85 %), Japan (7.56%), UK (5.11%), Malaysia (4.58%), Belgium (4.35%), Finland (2.95 %), Switzerland (2.02%), USA (1.91 %), Spain (1.35%), Bangladesh (1.31 %) and South Africa (1.21%) remained as major export destinations. About 91 % *asian* gum supplied to these top 12 destinations and rest was supplied to 23 countries. During 2017-18, Indonesia (85.82 %), Indonesia (7.81 %) and Singapore (6.37 %) have supplied *Asian* gum for India.

Table 27. Export and import of *Asian* gum during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
Germany	232.01	918.38	1420.00	50.31
Russia	45.00	161.63	250.00	8.85
Japan	21.60	138.06	210.00	7.56
U K	32.22	93.23	150.00	5.11
Malaysia	22.41	83.55	130.00	4.58
Belgium	26.40	79.43	120.00	4.35
Finland	18.00	53.81	80.00	2.95
Switzerland	10.00	36.95	60.00	2.02
U S A	14.32	34.80	50.00	1.91
Spain	4.20	24.68	40.00	1.35
Bangladesh Pr	12.44	24.00	40.00	1.31
South Africa	6.66	22.02	30.00	1.21
Hungary	6.00	21.24	30.00	1.16
France	10.00	19.52	30.00	1.07
Others (21)	37.11	114.07	180.00	6.26
Total	**498.37**	**1825.37**	**2820.00**	**100.00**
Import				
Korea RP	12.26	87.94	140.00	85.82
Indonesia	15.00	8.00	10.00	7.81
Singapore	8.98	6.53	10.00	6.37
Total	**36.24**	**102.47**	**160.00**	**100.00**

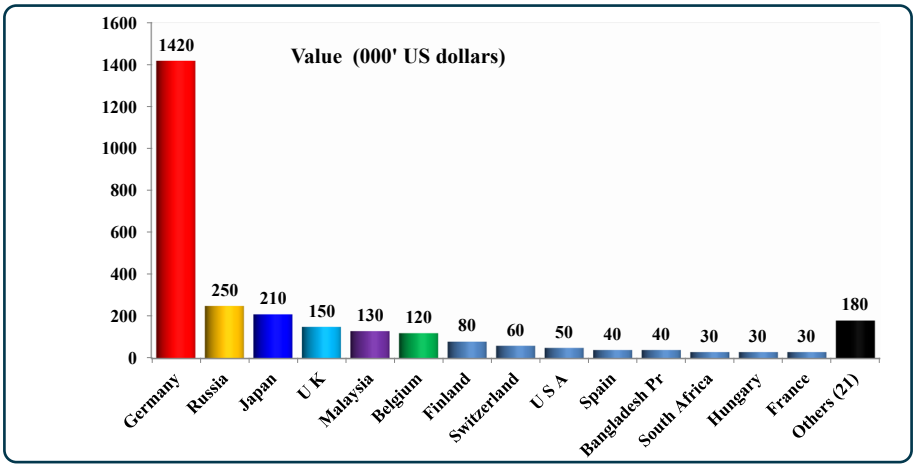


Figure 31. Destination wise scenario of *Asian* gum during 2017-18

International trade of *African* gum

In India, *African* gum is supplied from African countries for domestic consumption and rest is re-exported. It is evident from the Table 28 that France was only the major export destination of *African* gum. Import of *African* gum during 2017-18 is nil and export quantity of 0.77 ton may be from the carry over stock of previous years.

Table 28. Export of *African* gum during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US$)	Share (%)
France	0.10	3.29	10.00	76.51
Malaysia	0.67	1.01	5.00	23.49
Total	**0.77**	**4.30**	**15.00**	**100.00**

Tragacanth (Adracanth) *(Astragalus gummifer)*

Tragacanth gum (TG) is a natural and acidic polysaccharide that secretes spontaneously or with a scratch on the different species of *Astragalus* plant. It is also used as an adhesive in the rolling of cigars. EXIM scenario of Tragacanth (Adracanth) during 2017-18 is given in Table 29.

Table 29. Export of Tragacanth (Adracanth) during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
U K	4.50	50.12	80.00	57.35
Myanmar	12.00	34.33	50.00	39.28
Vietnam Soc Rep	4.00	2.04	5.00	2.33
Others (4)	0.35	0.91	5.00	1.04
Total	20.85	87.40	140.00	100.00
Germany	0.39	13.23	20.00	100.00
Total	**0.39**	**13.23**	**20.00**	**100.00**

Xanthan gum

Xanthan gum is a polysaccharide used as a food additive and rheology modifier, commonly used as a food thickening agent and in oil field drilling. The bacterium forms a slimy substance that acts as a natural stabilizer or thickener. In foods, *xanthan* gum is most often found in salad dressings and sauces. It helps to prevent oil separation by stabilizing the emulsion, although it is not an emulsifier. Toothpaste often contains *xanthan* gum, wherein it serves as a binder to keep the product uniform. EXIM scenario of *xanthan* gum during 2017-18 is given in Table 30.

Table 30. Export and import of *xanthun* gum during 2017-18

Country	Quantity(tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
U Arab Emts	1.56	8.53	9.72	65.21
Nigeria	1.35	3.39	8.16	25.92
Others (5)	0.37	1.16	2.12	8.87
Total	**3.28**	**13.08**	**20.00**	**100.00**
Import				
China P Rp	41.50	56.39	90.00	94.73
Poland	0.30	3.10	1.00	5.21
Sri Lanka Dsr	0.01	0.04	0.10	0.06
Total	**41.81**	**59.53**	**91.10**	**100.00**

Plant Exudate: Other gums

The other important gums are Gum kondagogu (*Cochlospermum gossypium), saja* (*T. tomentosa*), *Jhingan* (*L. coromandelica), palas* (*B. monosperma*), *khair* (*A. catechu*) and *char* (*B. lanzan* Spreng). Gujarat is major *guggal* gum and *prosopis* gum/*babool* gum producing state. Gum kondagogu is a naturally occurring nontoxic polysaccharide derived as an exudate from the bark of *C. gossypium* (*Bixaceae* family), a native tree of India. The total production of other gums across various states was around 80.0 tons during 2017-2018 (Table 31). It has been presented graphically in Figure 32.

Table 31. Production of other gums in India (in tons)

State /Year	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	Average
Andhra Pradesh	6.59	9.23	12.94	13.79	9.12	8.93	5.18	3.43	8.65
Telangana	1.16	2.24	0.60	0.65	3.50	3.55	5.00	2.00	2.34
Chhattisgarh	1.07	3.44	1.90	0.50	0.00	92.00	10.00	2.00	13.86
Gujarat	27.49	31.04	41.12	53.22	50.00	15.00	15.00	5.00	29.73
Jharkhand	240.60	200.50	125.60	60.20	40.00	50.00	60.00	15.00	98.99
Madhya Pradesh	280.00	250.00	200.00	174.60	100.00	100.00	100.00	20.00	153.08
Maharashtra	200.00	140.00	100.00	192.00	88.00	80.00	50.00	10.00	107.50
Others	100.00	80.00	70.00	72.30	42.00	40.00	40.00	20.00	58.04
Total	**856.91**	**716.45**	**552.16**	**567.26**	**332.62**	**389.48**	**285.18**	**77.43**	**472.19**

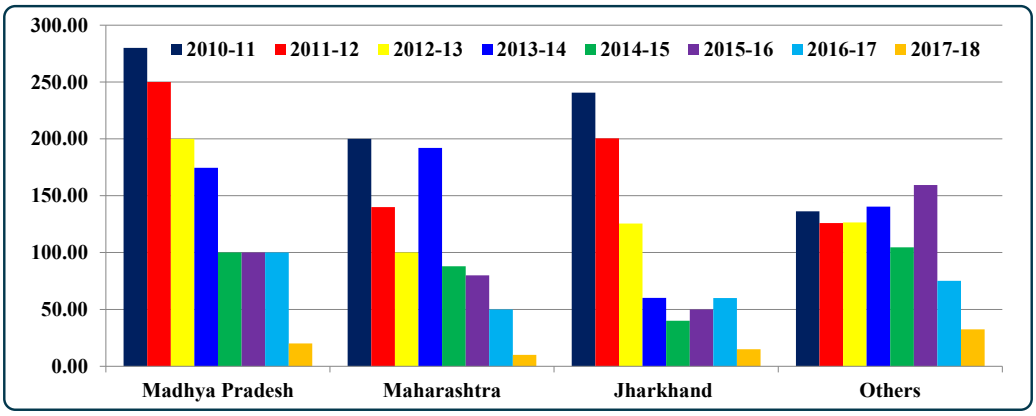


Figure 32. Total production of other gums across various states in India

Jhingan gum (Lannea coromandelica)

L. coromandelica (*Jhingan*, Indian Ash Tree, *Moi* tree) is a deciduous tree which is the source of *Jhingan* gum. It is mostly obtained from natural exudation and sometimes by tapping and annual gum production is about 50-60 tons in India. It is used as an alternative to gum *arabic* in food and other purposes.

Saja (Terminalia tomentosa)

T. tomentosa (*Saja, Saj*) is a tree growing upto 30 m tall, with a trunk upto diameter of 1 m. A copious transparent gum exudes in large globular tears from the trunk. In India, the annual *saja* gum production is about 30-40 tons. It is used as an incense and cosmetic. *T. tomentosa* has a remarkable attribute as some members of the species store water in the dry season.

Collection price and grading

The collection rate of *jhingan*, *khair* and *babool* gum ranged from ₹ 2000 to 5000/qt. Retail packing price of the *babool* gum was ranged about ₹200-220/kg. Currently, gum-producing *Acacia* species are harvested in Mauritania, Senegal, Mali, Burkina Faso, Niger, Nigeria, Chad, Cameroon, Sudan, Somalia, Ethiopia, Kenya and Tanzania. Grade-wise collection price of other gums in India is depicted in Table 32.

Table 32. Collection price of other gums in India (grade wise)

Natural gum	Scientific name	Grade	Price (₹/Kg)
Guggul gum	C. wightii	I	650-700
Guggul gum	C. wightii	II	250-275
Guggul gum	C. wightii	III	150-175
Salai gum	B. serrata	I	125-150
Khair gum	A. catechu	I	50-100
Babool gum	A. nilotica	I	100-150
Babool gum	A. nilotica	II	50-100
Babool gum	A. nilotica	III	30-50
Other			40-50

Regulations and governance

On the basis of regulations and governance issues for trading of NWFPs including natural resins and gums, the Government has classified as nationalized items, monopoly items and non-nationalized items. State Government classifies NWFPs as nationalized items, which can only be traded / marketed by Forest Department and it may vary from state to state. Four major gums namely *karaya/kullu* gum (*S. urens*), gum ghatti/ *dhawada* gum (*A. latifolia*), *babool* gum (*A. nilotica*) and *khair* gum (*A. catechu*) are categorized under nationalized gums in some states. A trade wing of the Forest Department's trade division deals with its marketing. The forest Department gives the monopoly rights for value added processes and marketing of NWFPs to some agency under lease agreement, based on sustainable harvesting and economic value. Apart from nationalized and monopoly items, remaining NWFPs are open to all. Primary collectors or their associations must get permits from the Divisional Forest Officer (DFO) to transport and market the produce. The state-wise list of gums producing trees and respective categories is presented in Annexure IV.

Seed based natural gum-*Guar* gum

The *guar* industry has evolved from being used as green pods for vegetable by households, whole green plant for cattle food and seed based for *churi* and *korma* used for domestic and ruminant feed purpose to find applications in industries. Owing to new technology, research and development the natural gum property of guar has varied applications from food, pharma industry to oil Industry too and is poised to grow. India is a native of *guar* or cluster bean *(Cyamopsis tetragonoloba)* grown in dry regions as rainfed crop and used as a green manure crop in agriculture. In major *guar* producing areas like Rajasthan, Haryana and Gujarat, *guar* crop is grown after the first shower of monsoon and is harvested in October-November.

Guar seed production in India

Guar gum comes from the endosperm (30%) of the seed. There are various grades of *guar* gums pure and its derivatives. *Guar* gum is a white to creamy colored, free flowing powder without any extraneous matter. Its ability to bind water, control the viscosity of aqueous solutions, form strong and tough films have accounted for its wider application and consequently the rapid growth.

India contributes about 80% to the world *guar* production and Pakistan is the second largest producer. All India area, production and yield of *guar* seed has been presented in Table 33. State wise area, production and yield of *guar* seed has been presented in Table 34.

Rajasthan has been a leading state in production as well as processing of *guar* seed in India followed by Haryana and Gujarat. In Rajasthan, Churu, Bikaner, Jaisalmer, Barmer, Nagaur, Hanuman Garh, Jodhpur, Ganganagar, Jaipur, Sirohi, Dausa, Jhunjhunu and Sikar are the districts where *guar* crop cultivated. Gurgaon, Mahendragrh and Rewari districts of Haryana and few districts like Kutch, Banaskantha, Mehsana, Sabarkantha, Vadodara and Ahmadabad of Gujarat are also *guar* producing areas in the country. Cultivation of *guar* crop has been started in newer areas like Maharashtra, Andhra Pradesh including Telengana and other southern states also.

Table 33. All India area, production and yield of *guar* seed and potential *of guar* gum production

(Area- thousand ha, Production- thousand tons, Yield- kg/ha,)

Year	Area	Production	Yield	Potential of *guar* gum production
2010-11	3387.9	1966.5	580	655.5
2011-12	3465.0	2233.1	644	744.4
2012-13	5156.9	2457.2	476	819.1
2013-14	5863.7	3390.7	578	1130.2
2014-15	5360.9	3285.0	613	1095.0
2015-16	5574.0	2815.2	505	938.4
2016-17	4060.0	1805.2	445	601.7
2017-18	3926.3	1916.6	488	639.9

Source: *Directorate of Economics and Statistics, MoA, GoI; www.rsamb.rajasthan.gov.in*

Table 34. State wise area, production and yield of *guar* seed during previous eight years

State	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	Average
State wise area of *guar* seed during previous eight years (000' ha)									
Andhra Pradesh	-	-	-	3.04	1.07	1.00	1.00	2.00	1.62
Gujarat	124.60	128.30	207.10	278.00	278.00	297.00	192.00	205.00	213.75
Haryana	256.00	215.00	388.00	481.00	406.00	457.00	315.00	273.00	348.88
Maharashtra	-	-	-	-	-	-	-	2.00	2.00
Punjab	24.00	25.00	26.00	28.20	48.60	30.00	20.00	10.00	26.48
Rajasthan	2980.95	3094.16	4533.27	5070.92	4625.21	4787.01	3530.01	3432.29	4006.73
Uttar Pradesh	2.40	2.50	2.50	2.50	2.00	2.00	2.00	2.00	2.24
All India	**3387.95**	**3464.96**	**5156.87**	**5863.66**	**5360.87**	**5574.01**	**4060.01**	**3926.29**	**4599.33**
State wise production of *guar* seed during previous eight years (000' tons)									
Andhra Pradesh	-	-	-	3.00	1.00	0.90	0.80	1.75	1.49
Gujarat	73.00	74.50	128.50	133.60	167.00	179.00	132.00	136.00	127.95
Haryana	333.00	290.00	278.00	369.00	338.00	380.00	248.00	223.00	307.38
Maharashtra	-	-	-	-	-	-	-	1.75	1.75
Punjab	18.00	20.00	22.00	21.20	33.40	30.00	18.00	8.00	21.33
Rajasthan	1540.54	1846.61	2026.67	2861.94	2743.97	2223.71	1404.82	1544.53	2024.10
Uttar Pradesh	2.00	2.00	2.00	2.00	1.60	1.60	1.60	1.60	1.80
All India	**1966.54**	**2233.11**	**2457.17**	**3390.74**	**3284.97**	**2815.21**	**1805.22**	**1916.63**	**2483.70**

State	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	Average
State wise yield of *guar* seed during previous eigh years (Kg/ hectare)									
Andhra Pradesh	-	-	-	987	937	900	800	875	900
Gujarat	586	581	620	481	601	603	688	663	603
Haryana	1301	1349	716	767	833	832	787	817	925
Maharashtra	-	-	-	-	-	-	-	875	875
Punjab	750	800	846	752	687	1000	900	800	817
Rajasthan	517	597	447	564	593	465	398	450	504
Uttar Pradesh	833	800	800	800	800	800	800	800	804
All India	**580**	**644**	**476**	**578**	**613**	**505**	**445**	**488**	**541**

Source: *Directorate of Economics and Statistics, MoA, GoI.*

Guar seed processing and value addition in India

India is leading in *guar* gum production due to its well established *guar* gum industry in Jodhpur, Rajasthan. Other states like Gujarat, Haryana and Maharashtra has *guar* seed collection centers and *guar* gum processing units. Recently, more units are also being established in Bikaner, Rajasthan with high capacity. Sri Ganganagar, Hissar, Alwar, Sirsa, Jodhpur, Bikaner, Jaipur are the major markets for *guar* seed. Mainly *guar* gum is used in paper, textile, oil drilling, mining, explosives, ore flotation and other various industrial applications.

Processing of *guar* gum are intended to impart desired chemical properties in the galactomannan like increased solubility in water, clarity of solution, increased shelf life, ionic character according to the needs of the particular final application. By these modifications lot of opportunities are opened, where chemically modified *guar* derivatives can be used (Table 35). Out of the decadal average potential production of *guar* gum around 6.4 lakh tons in the country, around 2.7 lakh tons is consumed in the domestic market as raw as well as processed and around 3.7 lakh tons is exported (Table 37).

Table 35. Various varieties of processed products of *guar* gum for industrial applications

Varieties	Applications	Varieties	Applications
Hydroxy alkylated *guar* gum	Oil well drilling	Sulphated *guar* gum	Mining
Carboxy methylated *guar* gum	Textile printing, Tobacco industry, water based paints	*Guar* gum formate	Food-Human and animal, Miscellaneous
Oxidised *guar* gum	Paper industry Food, textile printing applications	*Guar* gum acryl amide	Pharmaceutical industry
Acetates of *guar* gum	Food-Human and animal	Borate cross linked *guar* gum	Photography, Oil well drilling, carpet printing
Cationic derivatives of *guar* gum	cosmetic and paper industries	Reticulated *guar* gum	Stick explosives, blasting slurries

Movement in price of *guar* gum

The seasonal demand for *guar* gum is at its peak during the months of Nov-March around the world.

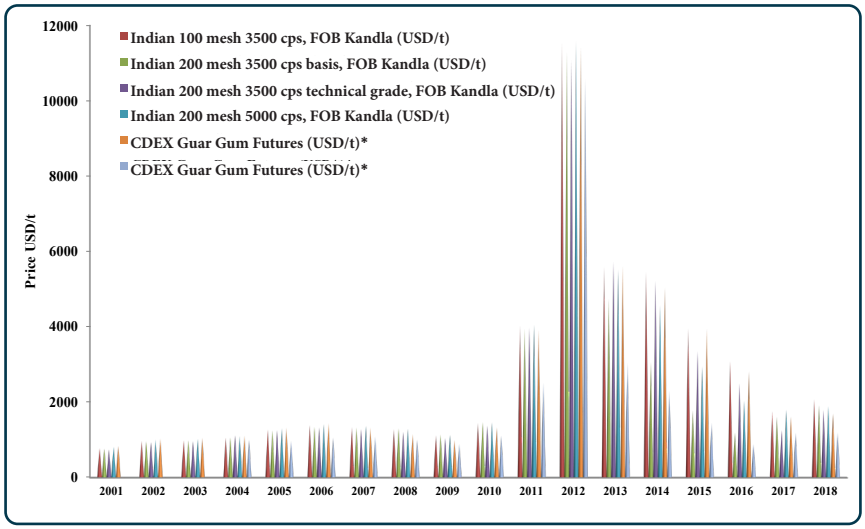


Figure 33. International price level of different grades of *guar* gum

As the reason being quite simple, the majority of the drilling activities of crude oils take place during these months. *Guar* seed prices in near future in domestic market may remain stable as normal production during 2017-18. The international price scenario of different grades of *guar* gum during 2001 to 2018 is shown in Figure 33.

International trade of *guar* gum

The share of agricultural exports in India's total exports increased from 9.71% in 2010-11 to 12.86% in 2017-18. Agricultural exports increased from ₹2396.81 billion in 2010-11 to ₹1110.18 billion in financial year 2017-18. Increase in the value of agricultural exports during 2017-18 was primarily on account of higher exports of wheat, other cereals, oil meals, *guar* gum, *basmati* rice and buffalo meat. India's top export of processed agricultural commodities in terms of quantity and value for three years is given in the Table 36. *Guar* gum is major export commodity with significant share of 2.33% in total agri-export during 2017-18.

India covers 53.9% world trade of *guar* gum followed by Spain (13.05%), Italy (9.01%), U S A (4.31%) and Pakistan Ir (3.54%). U S A (53.8%), Norway (6.7%), Russia (6.3%), China P Rp (6.0%) and Germany (4.9%) remained *guar* gum major importing Countries (*Source: UNCOMTRADE, as reported by the Importing countries*).

Export of *guar* gum during XIth and XIIth plan period, export and production capacity during 2007-08 to 2017-18 is given in Table 37 and Figure 34. India is a leading net exporter of *guar* seeds. During the year 2017-18, the country exports over 3.67 lakh tons of *guar* gum and its derivatives worth of ₹ 3715 crores (excluding guar meal) and it is comprised by 88% of treated and pulverized *guar* gum and 12% of refined split *guar* gum. Other *guar* based products were *guar* meal and traces of *guar* seed. Composition of *guar* gum export from

India is given in Figure 35. Destination wise overseas demand and source wise supply of *guar* meal, *guar* gum refined split (GGRS), *guar* gum treated and pulverized (GGTP) and in India from the overseas countries is illustrated through pie charts in Figure 36, Figure 37, Figure 38 and Figure 39, respectively.

Table 36. Composition of the agri-export commodities during previous three years

Particulars	Export Value (₹ in crores)			% share in total agri- export		
	2015-16	2016-17	2017-18	2015-16	2016-17	2017-18
Rice-basmati	22718.60	21512.91	26870.67	14.15	13.07	15.01
Buffalo meat	26684.22	26161.38	26035.19	16.62	15.89	14.54
Rice (Other than basmati)	15483.39	16929.88	23437.23	9.64	10.29	13.09
Spices	16630.15	19111.25	20084.91	10.36	11.61	11.22
Oil meals	3599.56	5410.10	7043.15	2.24	3.29	3.93
Castor oil	4616.10	4521.51	6730.00	2.87	2.75	3.76
Cashew	5027.99	5278.61	5945.28	3.13	3.21	3.32
Fresh vegetables	5237.10	5790.71	5297.72	3.26	3.52	2.96
Sugar	9824.52	8659.54	5225.60	6.12	5.26	2.92
Fresh fruits	4191.24	4974.21	4913.28	2.61	3.02	2.74
***Guar* gum**	**3233.87**	**3106.62**	**4169.56**	**2.01**	**1.89**	**2.33**
Processed fruits and juices	3766.99	3921.08	4169.13	2.35	2.38	2.33
Tobacco un-manufactured	4373.45	4249.85	3828.13	2.72	2.58	2.14
Cereal preparations	3358.11	3565.55	3561.69	2.09	2.17	1.99
Misc processed items	2907.81	3053.79	3548.95	1.81	1.86	1.98
Groundnut	4075.63	5444.33	3386.30	2.54	3.31	1.89
Sesame seeds	3012.31	2695.84	2990.93	1.88	1.64	1.67
Ayush and herbal products	2385.49	2693.57	2940.06	1.49	1.64	1.64
Tobacco manufactured	2078.91	2174.12	2193.58	1.29	1.32	1.22
Alcoholic beverages	2030.92	2004.79	2105.78	1.26	1.22	1.18
Dairy products	1677.32	1701.18	1954.63	1.04	1.03	1.09
Processed vegetables	1697.21	1765.75	1823.36	1.06	1.07	1.02
Other cereals	1702.50	1425.77	1604.28	1.06	0.87	0.90
Pulses	1656.03	1277.70	1469.63	1.03	0.78	0.82
Lac & its value added products	**203.31**	**225.53**	**285.18**	**0.13**	**0.14**	**0.16**
Others (15 commodities)	8399.48	6939.22	7455.89	5.23	4.22	4.16
Agriculture & allied products	**160572.21**	**164594.79**	**179070.11**	**100.00**	**100.00**	**100.00**

Source: *MoC&I, GOI, DGCIS, APEDA Annual Report & Indiastat.com*

Table 37. Export of *guar* gum during XIth and XIIth plan period

Year	Quantity (in '000 tons)	Value (in ₹ crore)	Value (000 US $)
11th Plan Period			
2007-08	124	611	276.63
2008-09	258	1334	284.95
2009-10	218	1130	227.26
2010-11	364	2749	631.65
2011-12	627	16407	3330.74
Average (Eleventh Plan)	**318**	**4446**	**950.25**
12th Plan Period			
2012-13	331	21147	3893.53
2013-14	469	11445	1931.60
2014-15	521	9153	1498.54
2015-16	256	3012	462.61
2016-17	318	2736	408.16
Average (Twelfth Plan)	**379**	**9499**	**1638.89**
2017-18	367	3715	576.35
Pooled average (XI & XII)	**349**	**6972**	**1294.57**

Source: MoC&I, GoI; APEDA Annual Report.

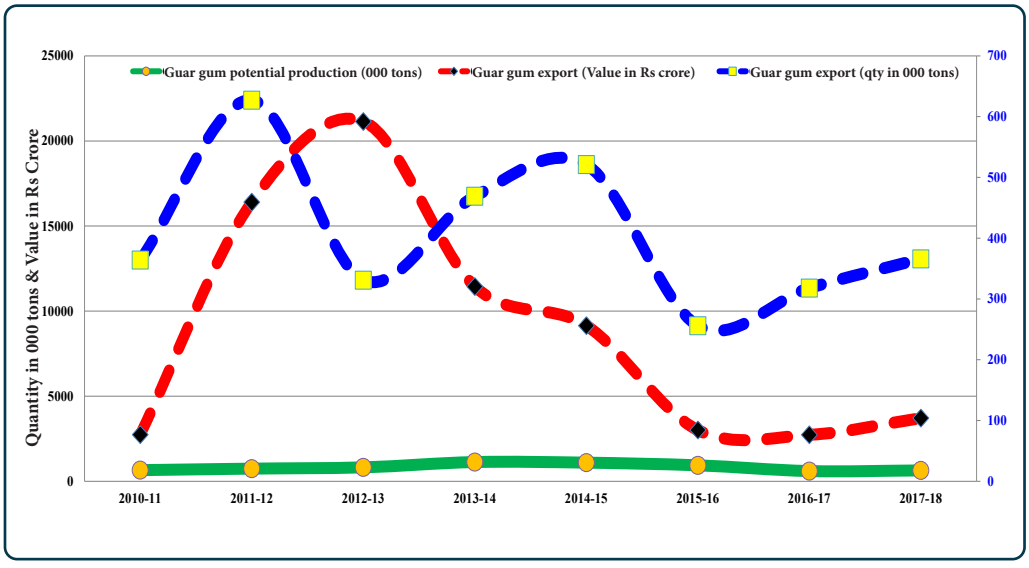


Figure 34. Trends in production potential and export of *guar* gum from India

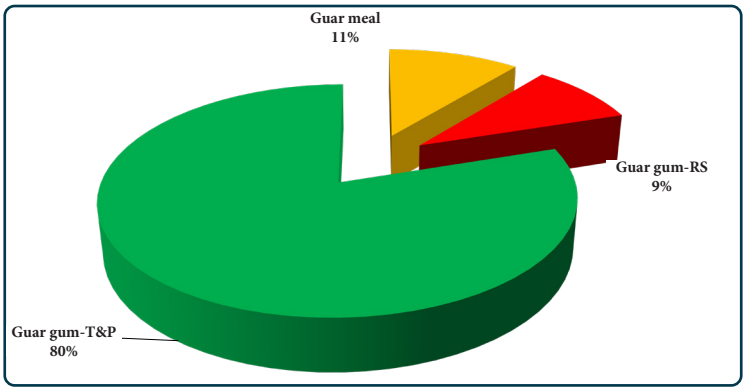


Figure 35. Composition of *guar* gum export from India during 2017-18

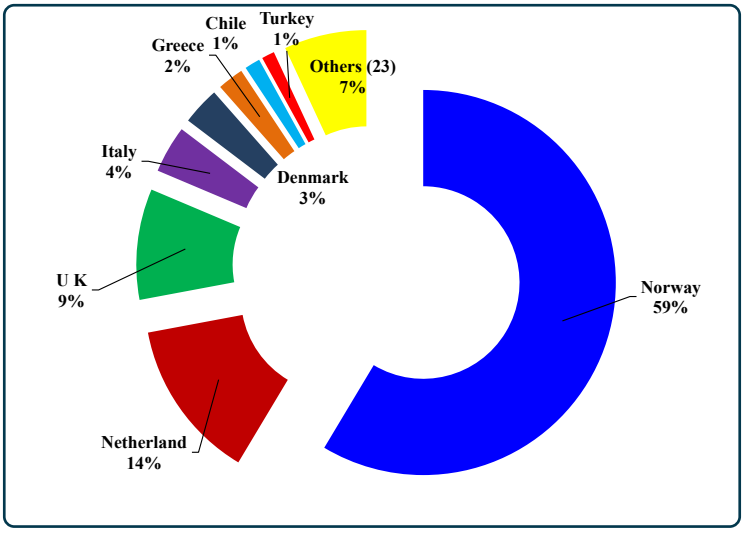


Figure 36. Destination wise scenario of *guar* meal during 2017-18

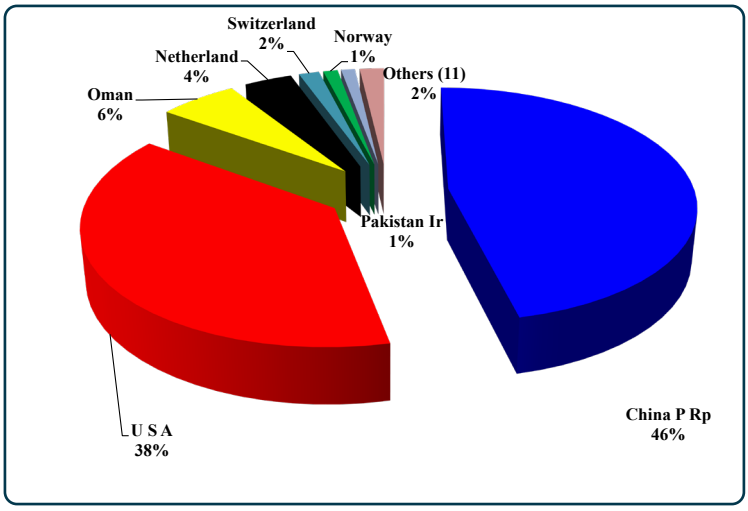


Figure 37. Destination wise scenario of *guar* gum refined split during 20167-18

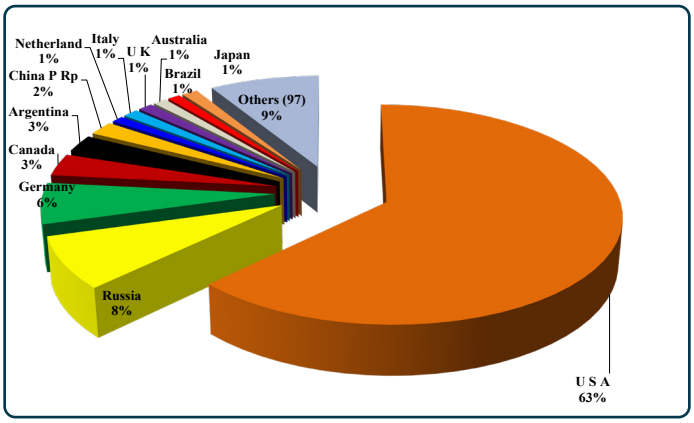


Figure 38. Destination wise scenario of *guar* gum treated and pulverized during 2017-18

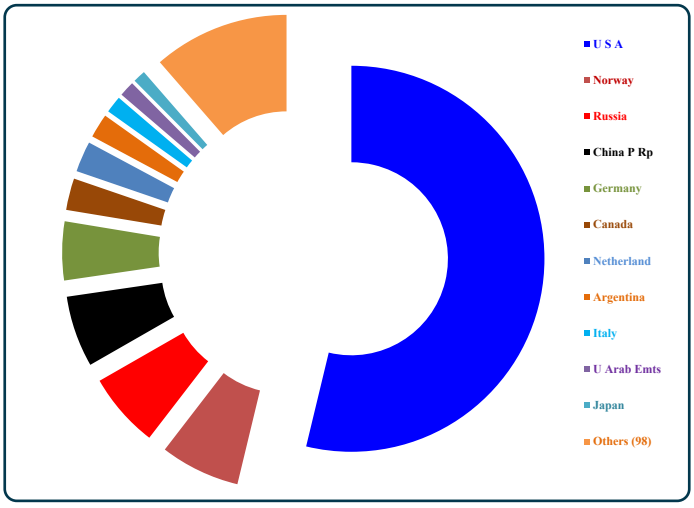


Figure 39. Destination-wise quantitative share of the *guar* gum exported from India (Overall)

United States, Norway, Russia, China, Germany, Canada, Netherlands, Argentina, Italy, UAE and Japan remained as major export destinations in 2017-18. During 2017-18, *guar* gum was exported across the globe to 109 countries. About 90 % was exported to top 11 countries and rest 10 % was exported to 98 countries around the world. Other countries are U K, Australia, Brazil, Indonesia, Oman, Mexico, Denmark, France, Thailand, Chile, Belgium, South Africa, Switzerland, Poland, Malaysia, Turkey, Greece, Korea RP, Spain, Iran, Saudi Arab, Egypt, Tunisia, Philippines, Israel, Pakistan, Latvia, Sri Lanka, *etc.* Destination wise export share is depicted in Figure 39.

Cassia gum

Cassia tora seeds, traded under ITC (HS) Code 09109915, have remained another commodity in high exports over the ten year period from 2005-06 to 2017-18 with its export volume increasing over ten times from 2083 tons during 2005-06 to 21218 tons during 2017-18, with a corresponding increase in export value from ₹ 3.57 crore in 2005-06 to ₹71.32 crore in 2017-18 (Table 38).

Table 38. Export and import of *Cassia tora* seeds during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
Vietnam Soc Rep	11910.20	2159.00	3350.00	30.27
Netherland	614.00	1106.97	1720.00	15.52
Taiwan	4870.00	926.12	1440.00	12.99
U K	637.00	637.83	990.00	8.94
Japan	1726.55	481.96	750.00	6.76
China P Rp	395.00	463.82	720.00	6.50
Russia	216.00	374.80	580.00	5.26
Australia	200.00	356.19	550.00	4.99
Thailand	66.00	184.85	290.00	2.59
Germany	155.00	91.77	140.00	1.29
Korea Rp	293.00	72.89	110.00	1.02
Others (12)	135.75	275.83	430.00	3.87
Total	**21218.50**	**7132.03**	**11070.00**	**100.00**
Import				
Unspecified	38.00	28.46	40.00	100.00
Total	**38.00**	**28.46**	**40.00**	**100.00**

Psyllium gum

Psyllium is the common name used for several members of the Isobgul plant genus *Plantago* whose seeds are used commercially for the production of mucilage. Seed produced from *P. ovata* is known in trading circles as white or blonde *psyllium*, Indian *Plantago* or Isobgul or Isabgul. Isobgul, the common name in India for *P. ovata*, comes from the Persian words ″isap″ and ″ghol″ that mean horse ear, which is descriptive of the shape of the seed. India possesses top position in *psyllium* gum production as well as in world export market.

Production Scenario of Isobgul and *Psyliium* gum

Area, production and yield indices of Isobgul increased over the decade in Rajasthan. Isobgul is an irrigated Rabi crop which remains in the field for about 4 months. It was a native of Persia, now grown in the western part of India. The main producing states are Rajasthan, Gujarat, Madhya Pradesh and Harayana as a cash crop. The main producing states are Rajasthan (Barmer, Nagaur & Jalore), Gujarat (Kuchchh & Ahmedabad) and Madhya Pradesh (10%). During 2017-18 season, overall sowing of isabgol in the country was nearly 5 lakh hectare, including over 4.17 lakh hectare in Rajasthan with a production of 2.26 lakh tons (40%), nearly 22,000 hectare in Madhya Pradesh with a production of 0.28 lakh tons and around 25,000 hectare in Gujarat with a production of 0.18 lakh tons. Overall production of Isobgul seed was almost 2.72 lakh tons during the season (Agricultural Statistics, 2017-18).

Price and Market Scenario of Isabgul

Pharmaceutical, Animal feed and Construction is the application sectors. The price of Isobgul is determined by visual inspection method. For the last 4 years approximately the price trend of average quality Isabgul seed was ₹20 –₹25/Kg. Three varieties usually arrive in the market- packed quality (premium quality), foreign quality and lower foreign quality.

Processing activities for *psyllium* gum

The processing activities are mainly undertaken in Gujarat. During the processing the millers remove *psyllium* husk from the Isobgul seeds. This process is called de-husking. The main byproducts are Lali (cattle feed), Chito (Pig feed) and Khakho (used to prevent ice slipping) and Golaisab (cattle feed).

Domestic and overseas demand

The local demand for the *psyllium* is also increasing day by day. About 20 to 25 % of total production of Husk is consumed in the domestic market. Indian pharmacy companies are using *psyllium* husk for making various drugs. There are various local brands having good market size across the country. The largest buyer of Isobgul from India is the Germany, USA, Sweden and Iran, accounting for around 75% of the total *psyllium* seed exports from India. The other main importers of *psyllium* husk and seeds are Australia, UK, Saudi Arab, Egypt, Canada, Poland, Brazil and South Africa (Table 39). Export of *Psyllium* husk was 39,708.36 tons during 2017-18. Thus, total international trade value of *psyllium* seed and husk from India was ₹1472.13 crore in 2017-18.

Table 39. Export of *psyllium* seed (Isobgul) during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Germany	458.25	621.15	960.00	43.78
U S A	84.09	277.58	430.00	19.57
Sweden	100.35	147.76	230.00	10.42
Iran	54.00	63.98	100.00	4.51
Australia	26.48	51.95	80.00	3.66
U K	22.48	38.50	60.00	2.71
Saudi Arab	12.32	34.83	50.00	2.46
Egypt A Rp	18.50	28.79	40.00	2.03
Canada	7.42	28.23	40.00	1.99
Poland	10.00	25.85	40.00	1.82
Brazil	20.50	23.51	40.00	1.66
South Africa	2.91	18.31	30.00	1.29
Others (27)	38.24	58.25	100.00	4.10
Total	**855.54**	**1418.69**	**2200.00**	**100.00**

Tamarind gum

Tamarind gum is obtained from endosperm of the seeds of the tamarind tree (*Tamarindus indica* linn). Tamarind gum powder has application in paper, textile industry, livestock feed, pet food, Mining & Pharmaceutical Industries *etc.* Export scenario in 2017-18 is depicted in Table 40.

Table 40. Export of flour meal and powder of tamarind during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
China P Rp	101.00	65.62	100.00	14.88
Germany	49.50	61.12	90.00	13.86
Sri Lanka Dsr	130.00	49.13	80.00	11.14
Australia	14.22	41.64	60.00	9.45
U S A	19.18	36.56	60.00	8.29
Indonesia	61.00	36.07	60.00	8.18
Netherland	60.00	28.90	50.00	6.56
U K	20.69	20.11	30.00	4.56
Switzerland	41.15	17.22	30.00	3.91
Egypt A Rp	23.00	14.62	20.00	3.32
Brazil	16.00	9.17	10.00	2.08
Canada	7.11	9.11	10.00	2.07
Korea Rp	13.00	8.06	10.00	1.83
Kuwait	2.70	7.49	10.00	1.70
Thailand	13.05	7.49	10.00	1.70
Japan	15.39	7.10	10.00	1.61
Others (21)	25.72	21.45	40.00	4.86
Total	**612.71**	**440.86**	**680.00**	**100.00**

C. NATURAL GUMS RESINS

Under this category, *asafoetida*, *salai* gum and *myrrh* are the major gum resins. About 1235 tons of the gum resins exported from the country and ₹615 million earned as a foreign exchange.

Asafoetida *(Hing)*

Asafoetida or asafetida is the dried latex prepared from the rhizome of several *Ferula species* occurring in Asia and North Africa (Kandziora & Melzig, 2015). Asafoetida (*F. anthrax* and *F. foetida*) is a popular spice used in daily food by Indians.

International trade of Asafetida

In India, Asafetida was supplied from Afghanistan (95%) and Uzbekistan (3%) during 2017-18. On other hand, UAE (24.84%), USA (15.88%), UK (6.88%), Singapore (6.34%), Myanmar (5.26%) and Saudi Arab (5.03%) are the major export destination of Asafetida in 2017-18 (Table 41).

Table 41. Export and import of asafetida during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
U Arab Emts	246.56	1316.50	2040.00	24.84
U S A	165.63	841.80	1310.00	15.88
U K	77.91	364.43	570.00	6.88
Singapore	61.44	335.83	520.00	6.34
Malaysia	61.46	278.79	430.00	5.26
Saudi Arab	69.51	266.63	410.00	5.03
Qatar	46.33	200.97	310.00	3.79
Australia	32.30	192.77	300.00	3.64
Thailand	37.49	160.09	250.00	3.02
Kuwait	26.49	155.95	240.00	2.94
Canada	25.36	152.60	240.00	2.88
Myanmar	43.11	150.28	230.00	2.84
Oman	39.00	140.08	220.00	2.64
South Africa	20.73	93.10	140.00	1.76
Baharain Is	24.27	83.24	130.00	1.57
Somalia	13.63	58.69	90.00	1.11
Sri Lanka Dsr	18.23	57.06	90.00	1.08
Others (81)	110.74	450.97	700.00	8.50
Total	**1120.19**	**5299.78**	**8220.00**	**100.00**
Import				
Afghanistan Tis	1200.59	66590.05	103330.00	95.18
Uzbekistan	74.93	2091.51	3250.00	2.99
Iran	140.57	1248.27	1940.00	1.78
Kyrghyzstan	1.10	25.45	40.00	0.04
Kazakhstan	0.15	3.58	10.00	0.01
Total	**1417.34**	**69958.85**	**108560.00**	**100.00**

Salai gum

Burseraceae family thrives in arid regions in the horn of Africa, West Asia, Oman, Yemen and southern Arabia. In India, it is widely distributed in Western Himalayas, Madhya Pradesh, Gujarat, Odisha, Rajasthan, Bihar, and Maharashtra. *Salai* gum (*Boswellia serrata*) Indian *Frankincense*, also known as *Olibanum or* Salai guggal (Kundur), is an aromatic oleo-gum resin (mixture of gum, resin, and essential oil) obtained from the bark of trees belonging to the genera *Boswellia* and usually collected by small nomadic groups. In India, *S. robusta*, *A. latifolia*, *T. tomentosa*, *B. serrata, B. lanzan, A. catechu, etc.* are reported as the important tree species by Champion and Seth (1968) and these species exhibit local dominance. The name is derived from the Arab word "al Luban", which means milk and is references to the milky sap that, exudes from the tree upon incision. It is also an important component in cosmetic industry and it is widely marketed as a food supplement (Khan, *et al.*, 2009).

Production and marketing scenario

This gum is tapped from the incision made on the trunk of the tree which is then stored in specially made bamboo basket and converted into different grades of material according to flavor, color, shape and size. Harvesting frankincense is a time consuming process that begins in December, reaching a peak from March to May (Marshall 2003). The trees start producing resin when they are about 8 to 10 years old (Michie *et al.*, 1991). Retail packing price *salai dhoop* in Gujarat increased from ₹250/kg to ₹400/kg and retail price of *Boswellia serrata* extract ranges from ₹1000/kg to ₹1200/kg.

International trade of olibanum

In India, olibanum is supplied from Somalia (63.43%), U Arab Emts (26.20%) and Kenya (6.53%). During 2017-18, Germany (60.83%), Trinidad (13.81%), Mexico (11.15%), Belgium (2.80%), France (2.69%), UK (2.04%) and Malaysia (1.34%) are the major export destinations (Table 42).

Table 42. Export and import of olibanum during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
Germany	25.25	98.90	150.00	60.83
Trinidad	9.48	22.46	30.00	13.81
Mexico	8.00	18.13	30.00	11.15
Belgium	1.20	4.55	10.00	2.80
France	1.80	4.37	10.00	2.69
U K	4.01	3.32	10.00	2.04
Malaysia	3.50	2.18	5.00	1.34
Sri Lanka Dsr	2.30	1.95	3.00	1.20
Mauritius	1.00	1.71	1.00	1.05
Others (12)	3.44	5.01	1.00	3.09
Total	**59.98**	**162.58**	**250.00**	100.00
Import				
Somalia	52.63	295.34	460.00	63.43
U Arab Emts	19.35	121.99	190.00	26.20
Kenya	52.00	30.38	50.00	6.53
Benin	16.00	17.88	30.00	3.84
Total	**123.98**	**465.58**	**720.00**	100.00

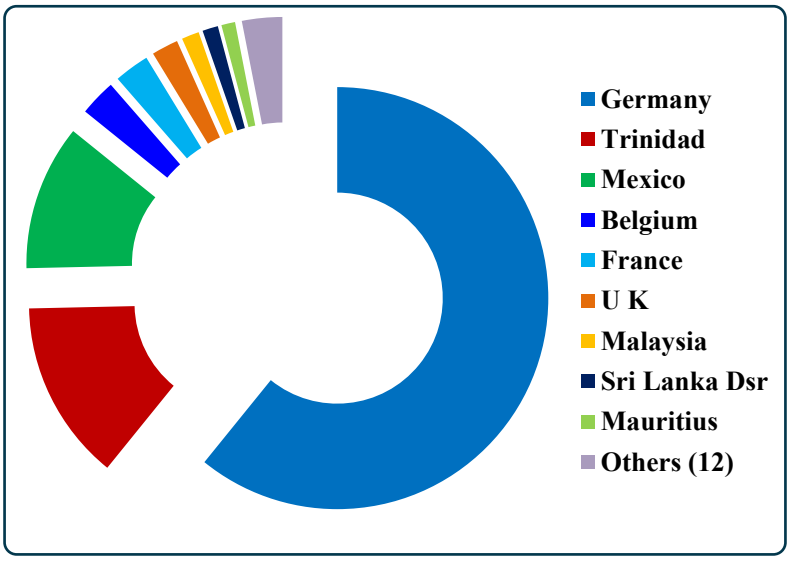


Figure 40. Destination-wise quantitative share of the olibanum exported from India

Myrrh

Myrrh is used raw or crushed and mixed with oil to make perfume. *Myrrh* comes from a small, thorny tree (*Commiphora myrrha).* The growers make a small cut in the bark, from where the resin would leak out. It is then collected and stored for about three months until it hardens into fragrant globules. Myrrh oil, which is steam distilled directly from the *myrrh* resin, has an aroma that is woody, earthy and a bit balsamic. Also, *myrrh* is occasionally used as a flavoring agent.

International trade of myrrh

In India, *myrrh* is mainly supplied from Somalia (99.28%), Iran (0.49%) and Italy (0.23%). During 2017-18, U Arab Emts (26.10%), UK (20.79%), Argentina (20.32%), Mexico (8.78%), Germany (7.85%) and USA (7.39%) are the major export destinations of *myrrh* (Table 43).

Table 43. Export and import of *myrrh* during 2017-18

Country	Quantity (tons)	Value (₹ lakh)	Value (000 US $)	Share (%)
Export				
U Arab Emts	0.10	1.13	6.30	26.10
U K	0.35	0.90	1.50	20.79
Argentina	0.50	0.88	1.50	20.32
Mexico	0.15	0.38	0.10	8.78
Germany	0.15	0.34	0.10	7.85
U S A	0.09	0.32	0.10	7.39
South Africa	0.03	0.22	0.10	5.08
Belgium	0.02	0.08	0.10	1.85
Hungary	0.05	0.06	0.10	1.39
Australia	0.00	0.02	0.10	0.45
Total	**1.44**	**4.33**	**10.00**	**100.00**
Import				
Somalia	39.46	326.51	510.00	99.28
Iran	2.50	1.62	0.10	0.49
Italy	0.10	0.76	0.10	0.23
Total	**42.06**	**328.89**	**510.20**	**100.00**

Dikamali (Gardenia gummiferra)

G.gummiferra (*Dikamali*) family Rubiaceae is the gum resin obtained from the leaf buds of a shrubby plant by making a cut on the stem or branches. It is geographically distributed in all districts of south India, Burma, Bangladesh, Konkan region, North Kanara, and Malabar Coast. Other names are *Dikamali, Gandharaj, Hingunadika, Nadihingu, pindava, etc.* It contains 89.9% resin and 0.1% oil and gardenin, a coloring agent and annual production is 10-15 tons. Producers' price in the market is ₹100-120/kg and retail price of processed product is about ₹1200-1400/kg.

D. NATURAL RESINS AND GUMS (NRGs)

NRG production level during 2017-18 was estimated to be comparatively higher (6.62 lakh tons) than previous year (Table 44). Other resins and gums production have increased during the current year. Production and trade of NRGs including *guar* gum, lac, pine resin, *karaya* gum, *dhawda* gum and other natural resins and total production figure of all the NRGs is tabulated as under.

Table 44.Total NRG production (quantity in tons)

Name of product	2013-14	2014-15	2015-16	2016-17	2017-18	Average	% share
Guar gum*	1130247.33	1094989.33	938403.67	601739.67	638877.33	880851.47	97.19
Lac	21008.00	16978.00	18746.00	16352.00	14230.00	17462.80	1.93
Pine resin	6875.30	6699.30	5726.10	5772.90	5254.40	6065.60	0.67
Karaya gum	128.76	83.23	100.35	145.00	150.00	121.47	0.01
Dhawda gum	448.41	295.09	194.00	240.00	200.00	275.50	0.03
Other gums	567.26	332.62	389.48	380.00	400.00	413.87	0.05
Other resins and gums	975.10	1189.76	980.00	1140.00	1500.00	1156.97	0.12
Grand Total	**1160250.16**	**1120567.33**	**964539.60**	**625769.57**	**660611.73**	**906347.68**	**100.00**

**Estimation of guar gum is based on the conversion of total guar seed production with a coefficient of conversion*

Export and import of Natural Resins and Gums

The total export of natural resins, gums and gum-resins during the year 2017-18 was 379979.77 tons valued ₹4064.96 crores and total import was 107399.56 tons valued ₹146878.70 crores. A comparative increase of 13.3% was observed in the volume of total NRG export during 2017-18 (Table 45 & Figure 41).

Table 45.Trends in export of NRGs and its share in total exports (value in Crores)

Year	Total Exports	Agri-cultural Exports	NRG Export	Share of Agri- exports in Total Exports (%)	Share of NRG Exports in Agri-exports (%)	Share of NRG Exports in Total Exports (%)
2010-11	1142921.92	111018.99	3086.23	9.71	2.78	0.27
2011-12	1465959.39	180528.60	16966.36	12.31	9.40	1.16
2012-13	1634318.84	223618.24	21761.19	13.68	9.73	1.33
2013-14	1905011.09	262778.96	12170.55	13.79	4.63	0.64
2014-15	1896445.47	239681.04	9632.71	12.64	4.02	0.51
2015-16	1716378.05	215396.55	3439.96	12.55	1.60	0.20
2016-17	1849433.55	226651.91	3108.13	12.26	1.37	0.17
2017-18	1956514.53	251563.94	4064.96	12.86	1.62	0.21
Average	**1695872.86**	**213904.78**	**9278.76**	**12.48**	**4.39**	**0.56**

A comparative increase of 13.3% was observed in the volume of total NRG export during 2017-18 (Table 45 & Figure 41). Due to improvement in price, particularly for *guar* gum, the total value of NRG export was higher (30.79%) than previous year. Details of export and import of natural resins, gums and gum-resins is presented in the Table 46.

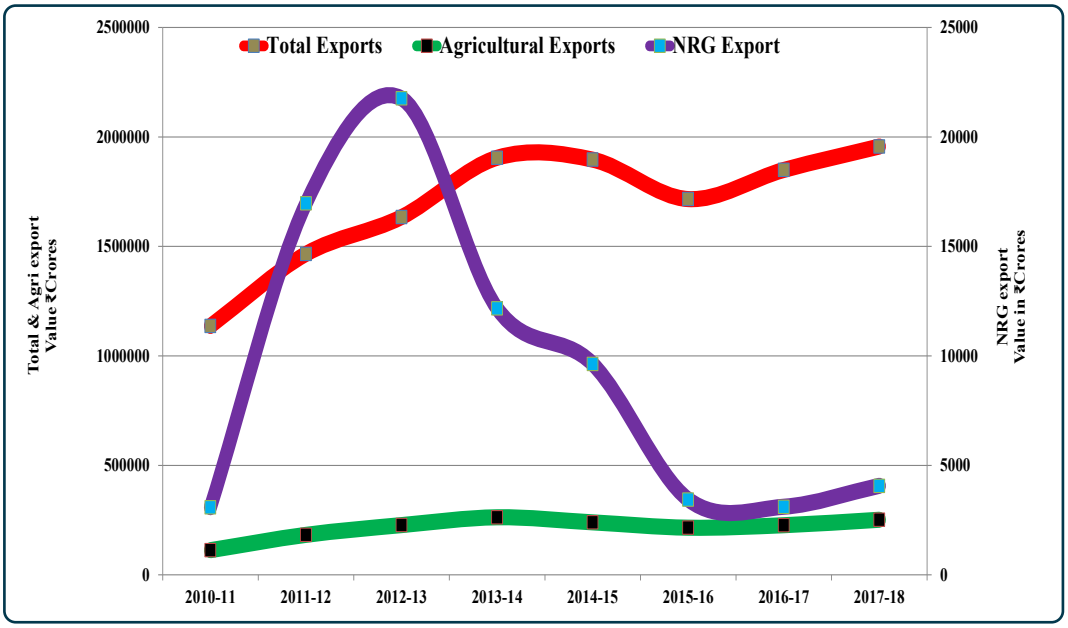


Figure 41. Trends in value of Total export, Agricultural export (Primary axis) and NRG exports (Secondary axis)

NRGs are important natural products in the export basket for foreign exchange reserve in India. The average annual export quantity of NRGs was 4.22 lakh tons valued ₹ 9278.76 crore during the period 2010-11 to 2017-18. However, during last 20 years exported quantity has registered increasing trend with an annual compound growth rate of 8.53%. Consequently the poor demand and sharp decline in *guar* gum prices, the export value of NRGs has also shown increasing trend with an annual compound growth rates of 16.42%.

Similarly, the average annual import quantity of NRGs was 0.91 lakh tons valued ₹ 1046 crores during the period 2010-11 to 2017-18. During last 20 years (1998-99 to 2017-18), import of NRGs had a very high stability with growth rates in import quantity (10.53%) as well as value (21.90%). Trend in export of natural resins and gums from India and import in India during last five years is presented in Figure 42 and Figure 43, respectively.

Table 46.Total NRG export from and import in India during 2017-18

Sl. No.	Name of product	Export		Import	
		Quantity (tons)	Value (₹ Lakh)	Quantity (tons)	Value (₹ Lakh)
A. Natural resins					
1.	Copal	12.18	7.18	2028.61	1622.40
2.	Dammar batu	35.44	33.33	20227.26	9144.44
3.	Mastic gum	0.00	0.00	2.20	11.59
4.	Gum rosin	288.42	294.26	46703.63	40835.86
5.	Lac	6889.48	22590.00	300.00	500.00
6.	Other resins	285.66	188.46	544.17	875.67
	Total	**7511.18**	**23113.23**	**69805.87**	**52989.96**
B. Gums					
7.	*African* gum	498.36	1825.37	36.23	102.47
8.	*Asian* gum	0.77	4.30	0.00	0.00
9.	Gum *arabic*	1814.55	1158.70	32075.52	16968.84
10.	*Guar* gum refined split	44740.39	37909.24	0.00	0.00
11.	*Guar* gum treated and pulverized	321923.78	333624.91	95.21	254.87
12.	Gum *Karaya* (Indian tragacanth)	313.40	815.76	2299.98	1990.76
13.	Tragacanth (Adracanth)	20.86	87.40	0.39	13.23
14.	Xanthum gum	3.27	13.08	41.81	59.53
15.	Other natural gums	1917.26	1789.23	622.42	1887.25
	Total	**371232.64**	**377227.99**	**35171.56**	**21276.95**
C. Gum resins					
16.	Asafoetida	1120.20	5299.77	1417.34	69958.85
17.	*Myrrh*	1.44	4.32	42.06	328.89
18.	Olibanum or frankincence	59.98	162.58	139.98	465.58
19.	Other gum resins	54.33	688.91	822.75	1858.47
	Total	**1235.95**	**6155.58**	**2422.13**	**72611.79**
	Grand Total	**379979.77**	**406496.80**	**107399.56**	**146878.70**

Source: *DGCIS, SHEFEXIL & APEDA Annual Reports.*

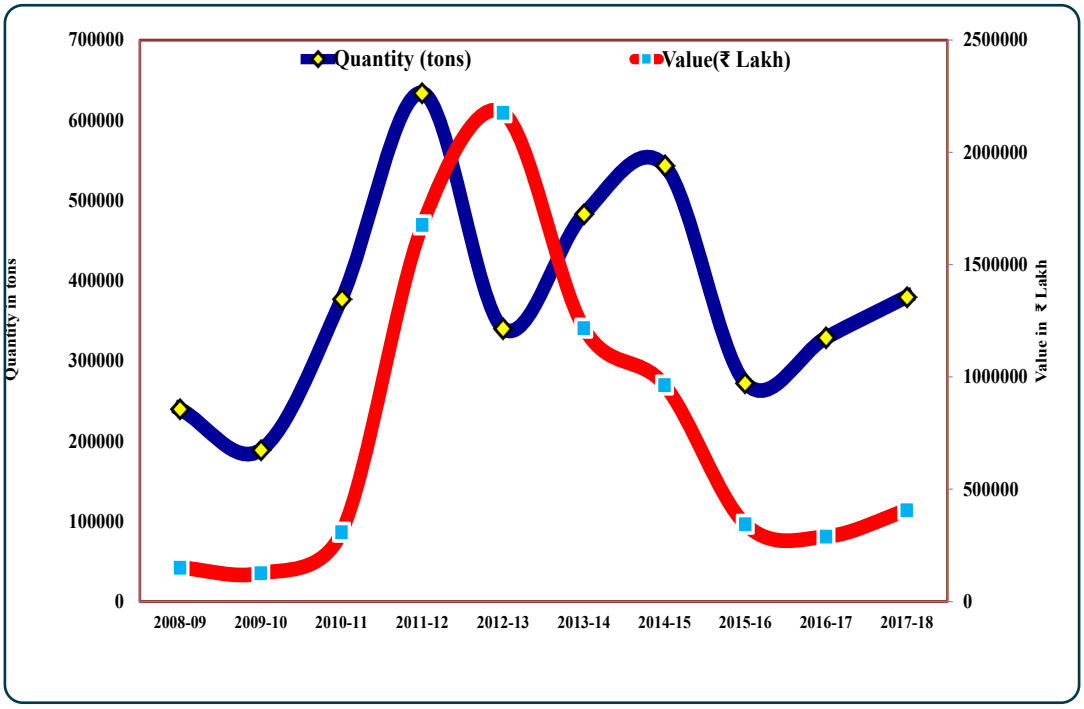


Figure 42. Trend in export of natural resins and gums from India during previous years

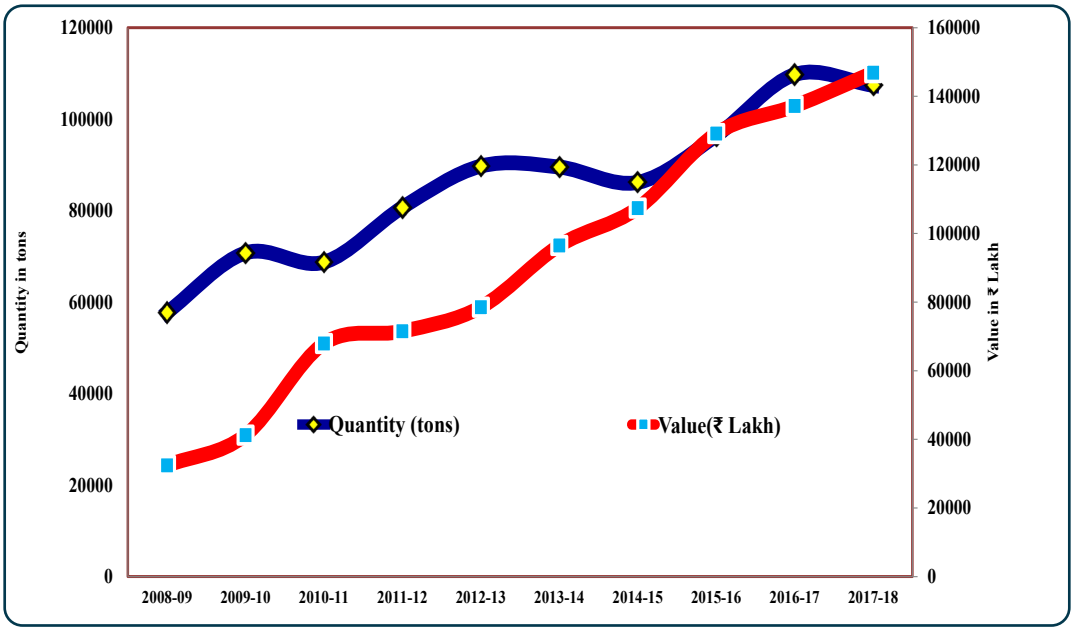


Figure 43. Trend in import of natural resins and gums in India during previous years

IV. POLICY INTERVENTIONS

According Tribal Cooperative Marketing Development Federation of India Limited (TRIFED), there are 3.65 lakh registered gatherers of NWFPs in India. To provide fair prices of NWFPs which differ widely state to state and to prevent exploitation of the poor tribals'; Minimum Support Price (MSP) was introduced. The practices adopted by different states for fixation of prices for different Non Wood Forest Products (NWFPs) were on the basis of market prices by reducing all the overheads and the second methodology for arriving at procurement price was to follow cost plus method in which cost of collection was needed to be calculated for NWFPs in different states. Therefore, government has taken the initiatives under the scheme for NWFPs.

A. Mechanism for marketing of Minor Forest Produce through Minimum Support Price and development of Value Chain for MFP

A Pricing Cell was constituted in pursuant to the Ministry of Tribal Affairs, Government of India and notified vide TRIFED, to recommend MSP for Minor Forest Produce under the scheme "Mechanism for marketing of Minor Forest Produce through MSP and development of Value Chain for MFP". To determine fair and remunerative MSP for lac and gum *karaya* the deliberations of the Pricing Cell with State Procurement Agencies (SPAs) and federations were conducted. Computation of cost of production of gum *karaya* and lac for MSP is being furnished by ICAR-Indian Institute of Natural Resins & Gums (ICAR-IINRG), Ranchi. The Pricing Cell analysed the above information furnished by the SPAs and observed the wide variation in the data on the related parameters from state to state. Following recommendations were made in the context of price stabilizing policy for NWFPs.

a) Minimum Support Price of gum *karaya* during 2014-15

Gum *karaya* is procured from gum pickers in three grades and procurement prices are also announced grade wise. Despite announcement of MSP in Gujarat hardly any quantity of gum *karaya* was procured. Similarly in Jharkhand and Maharashtra also, the quantity procured was very low. The Pricing Cell felt it appropriate to determine MSP on the basis of the cost of production as calculated by ICAR-IINRG plus 10% increase thereon. Accordingly, the Pricing Cell recommended MSP of ₹108/kg for gum *karaya* Grade-III for the current procurement season.

b) Minimum Support Price of lac for the year 2014-15

Major quantity of lac is available in the state of Jharkhand followed by Chhattisgarh, Madhya Pradesh and Odisha. Accordingly, based on the inputs provided by IINRG, Pricing Cell had recommended a Minimum Support Price (MSP) of ₹210 and ₹290/ kg for *rangeeni* and *kusumi* lac, respectively.

c) Minimum Support Price of lac for the year 2015-16

The Pricing Cell recommended enhancing the current MSP by 10%. Pricing Cell had recommended a MSP of ₹230 and ₹320/kg for *rangeeni* and *kusumi* lac, respectively.

d) Minimum Support Price of lac for the year 2016-17

During the current year, the lac growers have used broodlac of July 2015 crop in their cultivation when the prices were ranging between ₹150-200/kg, hence during harvesting season in December and January cost of production of lac may be comparatively lower than previous season. Pricing Cell had recommended a Minimum Support Price (MSP) of ₹100 and ₹150/kg for *rangeeni* and *kusumi* lac, respectively for 2016-17 crops.

e) Minimum Support Price of lac and gum *karaya* for the year 2017-18

During the current year, the lac growers have used broodlac of July crop in their cultivation when the prices were ranging between ₹150-220/kg, hence during harvesting season in December and January cost of production of lac may be comparatively lower than previous season. Pricing Cell had recommended a Minimum Support Price (MSP) of ₹100 and ₹150/kg for *rangeeni* and *kusumi* lac, respectively for 2017-18 crops. Further, committee advocated increasing the MSP. Recommended MSP for gum *karaya* was ₹108/kg during the current procurement season. Procurement of scrapedlac under the scheme is presented in Table 47.

Table 47. All India procurement of lac under Minimum Support Price (MSP) scheme

Year	Crop type	State Name/units	Chhattisgarh	Jharkhand	Gujarat	Total
2014-15	Lac *Rangeeni*	Quantity MT	0.00	0.00	0.00	**0.00**
		Value(₹ in Lakhs)	0.00	0.00	0.00	**0.00**
	Lac *Kusumi*	Quantity MT	0.00	0.00	19.51	**19.51**
		Value(₹ in Lakhs)	0.00	0.00	6.24	**6.24**
2015-16	Lac *Rangeeni*	Quantity MT	0.00	0.00	0.00	**0.00**
		Value(₹ in Lakhs)	0.00	0.00	0.00	**0.00**
	Lac *Kusumi*	Quantity MT	0.00	0.00	0.00	**0.00**
		Value(₹ in Lakhs)	0.00	0.00	0.00	**0.00**
2016-17	Lac *Rangeeni*	Quantity MT	101.17	0.00	0.00	**101.17**
		Value(₹ in Lakhs)	250.92	0.00	0.00	**250.92**
	Lac *Kusumi*	Quantity MT	350.63	3.91	2.66	**357.20**
		Value(₹ in Lakhs)	1208.14	5.87	0.46	**1214.47**
2017-18	Lac *Rangeeni*	Quantity MT	16.09	0.00	0.65	**16.74**
		Value(₹ in Lakhs)	17.71	0.00	1.02	**18.73**
	Lac *Kusumi*	Quantity MT	2.03	0.00	0.00	**2.03**
		Value(₹ in Lakhs)	3.05	0.00	0.00	**3.05**
Total		**Quantity MT**	**469.92**	**3.91**	**22.82**	**496.65**
		Value (₹ in Lakhs)	**1479.82**	**5.87**	**7.72**	**1493.41**

Source: *TRIFED, New Delhi*

B. Lac as an agriculture produce or forest produce?

The point whether lac is to be treated as an agriculture produce or forest produce has been raised at different fora. It varies from State to State around the government policy. But, lac growers, traders, processors and other stakeholders face the great problem in the spatial movement (transit) and processing of financial matters of this produce. They are not getting the benefits of priority sector from financial institution particularly for agriculture sector. Although, exporters of lac and its value added products are enjoying all the benefits under New Foreign Trade Policy for agriculture produce. Hence, it is an imperative step to be taken in the milieu of the million of lac growers and other stakeholders of this sector.

Lac as agriculture produce

I. Out of 25 crore households in India about one million farm families are engaged in lac cultivation across the country. Out of total 25 lakh households in Jharkhand state, there are about 3-4 lakh farm families/households engaged in lac cultivation and income from lac cultivation contributes about 25-32% (Table 48). Lac cultivation is labour intensive and eco-friendly activity.

II. All the cultural practices done by the farmers and his family members on their land like land preparation, seed sowing/nursery preparation, transplanting, manuring and use of fertilizers, pruning, inoculation of broodlac purchased by farmers from other lac growers/farmers/broodlac farms, *phunki* removal, spraying, intercultural operations and harvesting as well as sale of scraped lac in local market.

III. If it is cultivated on semialata plants, which is very similar to the any field crops at famers' field.

IV. Final produce at terminal market for export is considered as agricultural produce.

V. Agricultural and Village industry products presently covered under **VKGUY**. Lac based value added products have been covered under VKGUY in Foreign Trade Policy 2015-2020 and since long time these products considered under this category.

VI. Eco-friendly and green products that create wealth out of waste from agricultural and other waste products that generate additional income for the farmers, while improving the environment.

VII. It should be promoted among the farmers for supplementing their income and simultaneously women empowerment. It is labour intensive products with large employment potential and products with large number of producers and /or exporters, hi-tech products with high export earning potential.

Table 48. State wise distribution of lac growers

Particulars	No. of households (in lakh)		No. of members under Cooperative Federations/ SHGs
	As per new estimates	As per museum display	
	2017-18	2007-08	
Jharkhand	4.00	3.00	55000
Chhattisgarh	2.00	2.00	43391
Odisha	1.00	1.00	0
Maharashtra	0.75	0.75	0
Madhya Pradesh	0.75	0.75	0
Andhra Pradesh	0.50	0.50	0
West Bengal	0.30	0.30	8110
Telengana	0.20	0.20	0
Gujarat	0.20	0.20	1000
Assam	0.10	0.10	0
Meghalaya	0.05	0.05	0
Manipur	0.05	0.05	0
Arunachal Pradesh	0.05	0.05	0
Rajasthan	0.05	0.05	0
Others	0.10	0.10	1000
Total (All India)	**10.10**	**9.10**	**108501.00**

Lac as not a forestry produce

I. Practically all the lac produced in the country is cultivated at farmer's field by farmers, not a wild one. Major portion of the total lac in the country is produced by the farmers.

II. Only minor portion is collected by the forest dwellers during collection of other NWFPs.

III. In many states it has been categorized as non-nationalized item. It means there is no monopoly of the forest department to claim on its production and further marketing.

Table 49. Trainings conducted by ICAR-IINRG Ranchi

Extension activities	No. of stakeholders trained during 1995-96 to 2016-17
Educational Program (One week)	1254
Farmer's Training (One week)	11053
Kisan Mela	3312
Lac based product demonstration training	192
Model Training	50
On-farm Training (1-2 days)	39244
On-farm training Supplementary/ Motivational (One day)	38289
Orientation Program (One day)	39621
Short term training program (3 days)	349
Trainer's Training (One week)	3393
Total	**136757**

Due to continuous pursuance Ministry of Environment, Forests & Climate Change vide Gazette Notification No 43 dated 23 November, 2017 has amended section 2 of Indian Forest Act 16 of 1972, in clause (7) the word " bamboos" shall be omitted. Thus bamboo will not be considered as forest species; hence the transit regulations will not be applicable. Eight states relaxed the transit regulations during 2016-17 and 13 more states relaxed the transit regulations in 2017-18 to take forward the national initiative on growing green cover. To promote agro-forestry, a dedicated National Agro-forestry Policy was introduced in 2014. The Sub-Mission on Agro-forestry (SMAF) has been made operational from 2016-17 to promote "*Har Medh Par Ped* (Trees on Farm Bund)".

C. Unit cost for lac sector investment activities

To promote capital formation in farm sector, Government of Jharkhand emphasized on investment plans. Consequently, National Bank for Agriculture and Rural Development (NABARD) introduced the investment plans for lac sector in Jharkhand. Unit Cost of Farm Sector Investment Activities in Jharkhand for 2017-18 is depicted in Table 50.

Table 50. Unit cost of lac sector investment activities in Jharkhand 2017-18

Particulars	Unit size	Unit cost (In `)
Scientific Lac cultivation		
Kusmi Lac Cultivation on *Ber*	50 Tree	51000
Kusmi Lac Cultivation on *Kusum*	50 Tree	150000
Kusmi Lac Cultivation on *Semialata*	1000 plants (0.25 acre)	87000
Rangeeni Lac Cultivation on *Palas*	100 tree	54000
Lac processing & value addition		
Small Scale Lac Processing Unit	Capacity of 100kg seedlac/day	3100000

Source: *ICAR-IINRG Ranchi*

V. CONCLUSIONS

NRG production level in India during 2017-18 was comparatively higher (6.62 lakh tons) than previous year. Other resins and gums production have increased during the current year. Total production of the lac in the country was 14,230 tons which is lower than the previous year production (16,352 tons) comprising *rangeeni* (5484 tons) and *kusmi* (8746 tons) sticklac. It is interesting to mention that the production level of lac is sustained to 14,230 tons during 2017-18 which is about 57 % higher in comparison to the lowest level of 9,035 tons during 2010-11. Although, it is about 32% lower than the highest level of 21,008 tons in 2013-14. In comparison to previous year, the production of *ranggeni* as well as *kusmi* crop during 2017-18 decreased by 11.4% and 13.9%, respectively. The decrease in production was comprised of 14.3 % and 13.5 % fall in the production of *jethwi and aghani* crops, respectively. However, a decrease of 11.5% and 11.4% was observed in case of *baisakhi* and *katki* crops, respectively .Overall, the total lac production for the year 2017-18 has decreased by about 13 % in comparison to 2016-17. Among the lac growing states, Jharkhand state ranks 1st (54.6%) followed by Chhattisgarh (16.5%), Madhya Pradesh (13.9%), Maharashtra (6.0%), West Bengal (5.4%) and Odisha (2.2%). These six states contribute more than 98 % of the total lac production in India. Among the different cropping season crops, *aghani* crop was ranked 1st with the contribution of 34.4 % followed by *jethwi* (27.1 %), *baisakhi* (18.8 %) and *katki* (19.7 %) in total lac production.

About 80 % of natural gum production in the country is contributed by Maharashtra (37.3 %), Madhya Pradesh (18.4 %), Jharkhand (11.1 %), Telengana (6.6 %), and Chhattisgarh (5.7 %). Rest of the 20 % comes from Gujarat (4.2 %), Andhra Pradesh (3.4%) and other minor gum producing states (13.2%). India contributes about 80% to the world *guar* seed production and Pakistan is the second largest producer. During the year 2017-18, the country exports over 3.67 lakh tons of *guar* gum and its derivatives worth of ₹ 3715 crores (excluding guar meal) and it is comprised by 88% of treated and pulverized *guar* gum and 12% of refined split *guar* gum. Other *guar* based products were *guar* meal and traces of *guar* seed.

Major gums and resins of international commerce produced and traded in and from India include *guar* gum, lac, gum *karaya*, gum *arabic* from *Acacia* species, tamarind gum powder, psyllium gum,*Cassia tora* and *myrrh*, asafetida, rosin and olibanum from *Commiphora* species. The volume of export of natural resins increased significantly and the volume of import of natural resins was also decreased slightly in 2017-18. The share of agricultural exports in India's total exports increased from 9.71% in 2010-11 to 12.86% in 2017-18. Agricultural exports increased from ₹2396.81 billion in 2010-11 to ₹1110.18 billion in financial year 2017-18. Increase in the value of agricultural exports during 2017-18 was primarily on account of higher exports of wheat, other cereals, oil meals, *guar* gum, *basmati* rice and buffalo meat.

The total export of natural resins, gums and gum-resins during the year 2017-18 was 379979.77 tons valued ₹4064.96 crores and total import was 107399.56 tons valued ₹146878.70 crores. NRGs are important natural products in the export basket for foreign exchange reserve in India. The average annual export quantity of NRGs was 4.22 lakh tons valued ₹ 9278.76 crore during the period 2010-11 to 2017-18. Due to increase in price, particularly for *guar* gum, the total value of NRG export was higher (30.79%) than previous year. During last 20 years

exported quantity has registered increasing trend with an annual compound growth rate of 8.53%. Consequently the poor demand and sharp decline in *guar* gum prices, the export value of NRGs has also shown increasing trend with an annual compound growth rates of 16.42%. Similarly, the average annual import quantity of NRGs was 0.91 lakh tons valued ₹ 1046 crores during the period 2010-11 to 2017-18. During last 20 years (1998-99 to 2017-18), import of NRGs had a very high stability with growth rates in import quantity (10.53%) as well as value (21.90%).

Average annual export quantity of lac during last ten (2010-11 to 2017-18) years was 6,760.59 tons valued ₹32842.52 lakh. Average annual demand of lac was 6,799.60 tons valued ₹36515.19 lakh during the period 2012-13 to 2016-17. However, during last eight years (2010-11 to 2017-18) Annual compound growth rate in exported quantity was positive for lac (2.81%) and total natural resins group (1.67%) as the acceleration in quantity demanded since 2012-13 onwards. But, in value terms the same figure has shown decelerating growth rates. During 2017-18, exported value showed that 99.37 % contribution in total export of natural resins were from lac (91.72 %), gum rosin (3.84 %) and other resins (3.81 %). The total export of lac and its value added products during the year 2017-18 was 6889.48 tons which was valued ₹225.90 crores.

The analysis of EXIM data of rosin revealed that the exported quantity is increased by 102.2% while imported quantity of rosin was increased by 8.37% in 2017-18 in comparison to 2016-17. During this year in international market, export price of rosin decreased by 19.02 %, and import price decreased by 4.82 % comparatively of previous year. Demand of resin was higher than the quantity processed in India. Hence, India depends on Indonesia, Brazil, Nepal, China and Vietnam to meet its domestic demand. About 99 % of gum rosin was supplied from these five neighboring countries. About 1235 tons of the gum resins exported from the country and ₹615 million earned as a foreign exchange during 2017-18.

Out of 25 crore households in India about one million farm families are engaged in lac cultivation across the country. According to The Tribal Cooperative Marketing Development Federation of India Limited (TRIFED), there is 3.65 lakh registered gatherers of NWFPs in India. To provide fair prices of NWFPs which differ widely state to state and to prevent exploitation of the poor tribals'; Minimum Support Price (MSP) was introduced. Pricing Cell had recommended a Minimum Support Price (MSP) of ₹100 and ₹150/kg for *rangeeni* and *kusumi* lac, respectively for 2017-18 crops. Further, committee advocated increasing the MSP. Recommended MSP for gum *karaya* was ₹108/kg during the current procurement season.

Practically all the lac produced in the country is cultivated at farmer's field by farmers, not a wild one. Major portion of the total lac in the country is produced by the farmers. Therefore, lac is not a forest produce. To promote agro-forestry, a dedicated National Agro-forestry Policy was introduced in 2014. The Sub-Mission on Agro-forestry (SMAF) has been made operational from 2016-17 to promote "*Har Medh Par Ped* (Trees on Farm Bund)". To promote capital formation in farm sector, Government of Jharkhand emphasized on investment plans. Consequently, National Bank for Agriculture and Rural Development (NABARD) introduced the investment plans for lac sector in Jharkhand.

Under the capacity building and skill development programs, stakeholders got benefited

directly and farm level adoption was observed. Due to increase in awareness through skill development programs for the beneficiaries, it could be promising strategy for enabling them to maximize the utilization of available natural resources including lac host trees, trees for gum exudates, rainfed areas of western arid zones for *guar* crop and plateau region for *Cassia* tora and tamarind seeds. Strengthening of value chain through policy interventions like MSP and farmer-industry linkages through village level organizations will be helpful to harness the full potential of available resources. This may be helpful to generate gainful employment on the one hand for checking migration while also providing quality output to the hands of the consumer.

Value addition in lac based products has an encouraging market scenario as the export of Aleuritic acid is increasing since last two years. To meet the domestic demand pine resin processing infrastructure is needed to be revamped. Production and procurement of some gums is declining during last three years. MSPs for NWFP may be key intervention for procurement, marketing and value chain establishment. It will be helpful to stabilize the price fluctuation particularly in case of lac.

Enhancing the offer of Indian market products and increasing their role in the rural economy could help to reduce the costs of forest protection. A well-structured forest economy that is able to provide stable flow of incomes can provide a fundamental set of public non-market services and social values to both local people and the whole community. Understanding the true value of natural resources, for both land users and policymakers, is an essential step for promoting their protection and sustainable use. Since markets do not reflect all values flowing from NWFPs, further research is needed to better understand the link between ecosystem functioning and the delivery of services, as well as their appropriate economic evaluation.

The research gaps related to lac production can be addressed through structural, functional genomics and sap analysis for in vitro production of lac resin components; quick and easy establishment along with improved management strategies for lac host plants to increase production and productivity of lac; ex-situ conservation of lac insect and host biodiversity vis-a-vis documentation with digital illustrations of lac insect species and mapping of lac host tree; strategies to combat influence of climate change on lac insect life cycle and production; diversification of lac production cropping system for enhancing production to ensure food security; impacting the livelihood of the stakeholders through capacity building of farmers and entrepreneurship development in India; estimation of demand and supply for market decision support and price policy (MSP); Impact assessment & feedback of the stakeholders through linkages; policy interventions like lac as agricultural produce, capital formation through financial schemes for viable enterprises, establishing National Lac Development Board (NLDB), *etc.*

The research gaps related to processing of NRGs can be addressed through fortification of cereals, fruits, vegetables, spices etc. through coating / encapsulation; preparation of nutrient fortified biodegradable film / packaging material from different gums; product and application development and value addition; equipment and technology for replacement of traditional method of drying, processing, packaging & storage and waste management; and development of NRGs based value added product i.e functional food, food & pharmaceutical,

additives, prebiotics, nano-composite film and biodegradable packaging material. The reported resins offer huge potential as a possible pharmacological application but its necessary a further investigation to verify whether purified compounds isolated may have better therapeutic potential as compared to crude extracts. These new chemicals will serve to enhance the continuous usefulness of higher plants and their products as renewable resources of chemicals.

The natural gums and resins are safe to humans, biodegradable and eco-friendly. This sub-sector generates employment and income opportunity, particularly in off-agricultural season. The forest sector supports about 50 million population inhabiting forest and sub-forest areas and about 70% of employment in the sector is in minor forest produce, which includes natural resins and gums. India has a great potential in production and export of natural resins and gums because of availability of resins and gums yielding trees, manpower, favorable climate and R&D support. The proper attention and action for intensification of effort for increasing natural resins and gums production will definitely improve the export earning of the country.

Annexure I. Sticklac production scenario in India during 2017-18 (in tons)

Name of states / Districts	*Baisakhi*	*Katki*	*Jethwi*	*Aghani*	Total
Andhra Pradesh	5	3	1	0	9
Assam	5	154	0	0	9
Chhattisgarh					
Korba	180	140	110	200	630
Kanker	40	40	150	180	410
Bilaspur	50	65	20	40	175
Surguja	95	30	5	0	130
Mahasamund	50	50	10	15	125
Rajnandgaon	50	40	10	20	120
Bastar	35	5	25	40	105
Dhamtari	30	15	30	25	100
Raipur	5	2	30	60	97
Raigarh	25	40	10	20	95
Janjgir-Champa	30	10	10	20	70
Durg	10	10	0	0	20
Others	50	50	10	65	175
Total	650	497	420	685	2252
Gujarat	10	2	5	5	22
Jharkhand					
Ranchi	180	110	1150	1100	2540
Simdega	15	50	870	920	1855
Khunti	65	60	390	600	1115
Gumla	10	10	475	500	995
West Singhbhum	30	10	230	400	670
Latehar	8	10	1	80	99
Palamau	10	10	0	70	90
Saraikela Kharsanwa	10	5	20	45	80
Giridih	20	10	0	25	55
Chatra	5	0	0	45	50
Dumka	0	0	30	10	40
Bokaro	0	0	15	20	35
Hazaribag	0	0	0	30	30
Lohardaga	0	0	15	15	30
Jamtara	0	0	15	0	15
Dhanbad	5	0	5	0	10
Garhwa	5	2	0	0	7

Name of states / Districts	*Baisakhi*	*Katki*	*Jethwi*	*Aghani*	Total
Deoghar	0	0	5	0	5
East Singhbhum	0	0	1	1	2
Others	0	0	4	19	23
Total	363	277	3226	3880	7746
Madhya Pradesh					
Seoni	450	400	12	2	864
Balaghat	370	370	5	1	746
Mandla	170	50	5	15	240
Dindori	50	15	8	0.5	73.5
Hosangabad	25	20	5	0.5	50.5
Chhindwada	10	15	8	1	34
Narshinghpur	2	20	1	1	24
Annuppur & Shahdol	0.5	1	2	0	3.5
Others	110	60	10	20	200
Total	1187.5	951	56	41	2235.5
Maharashtra					
Gondia	290	370	0	0	660
Garhchiroli	70	30	2	8	110
Chandrapur	10	40	0	0	50
Bhandara	20	25	0	0	45
Total	390	465	2	8	865
Meghalaya (Garo hills)	5	1	0	0	6
Odisha					
Sundergarh	5	0	50	50	105
Nabarangpur	0.5	1	25	40	66.5
Mayurbhanj	0.5	1	6	40	47.5
Keonjhar	0.5	0	5	10	15.5
Koraput	0.5	1	10	0	11.5
Balasore	0.5	1	5	2	8.5
Others	5	5	10	5	25
Total	12.5	9	111	147	279.5
Uttar Pradesh	10	2	0	0	12
West Bengal					
Purulia	10	200	20	95	325
Midnapur	20	230	2	5	257
Bankura	5	50	2	5	62
Others	10	110	5	25	150
Total	45	590	29	130	794
Grand total	2683	2801	3850	4896	14230

Annexure II. Lac producing areas in the country

Sl. No.	Name of State/ District	Major lac producing areas
I.	**Andhra Pradesh**	
1	Adilabad	Utnoor
2	Vishakhapatnam	Paderu
II.	**Assam**	
3	Kamrup	Boko
4	Karbi-Anglong	Amtreng, Baithalangsu
5	Marigaon	Nellei
6	Nagaon	Amsoi, Hojai
III.	**Bihar**	
7	Gaya	Raniganj
8	Jehanabad	Malichak
IV.	**Chhattisgarh**	
9	Balrampur	Chando, Dharmi, Wandrafnagar
10	Bastar	Keshkal
11	Bilaspur	Gaurella, Kota, Lorami, Pendra
12	Dhamtari	Gatta Silli, Nagri, Sihawa
13	Durg	Balod, Daudi, Dondi-Lohara, Kusumkasa
14	Gariyaband	Amlipadar, Chhaila, Chhura Gariyaband, Indagaon, Mainpur, Udanti
15	Janjgir-Champa	Sakti, Saragaon
16	Kanker	Antagarh, Bhanupratapur, Biragaon, Kondagaon, Korar, Narharpur, Sambalpur
17	Korba	Bhaisama Bazar, Chaitama, Chhuri, Haldi Bazar, Kartala, Katghora, Korbi, Madanpur, Pali, Pasan, Pasarkhet
18	Korea	Bhartarpur, Kusmi
19	Mahasamund	Bagbahara, Basana, Khalari, Mahasamund, Pithora, Tendukona
20	Narayanpur	Narainpur
21	Raigarh	Dharmjaigarh, Lailunga, Pathalgaon,Tamnar
22	Raipur	Abhanpur
23	Rajnandgaon	Aawadhi, Bharitola, Khardi, Manpur, Mohala-chowki
24	Sarguja	Chalgi, Mainpat, Pasta, Pratappur, Premnagar, Raghunathnagar, Ramanujganj
V.	**Gujarat**	

Sl. No.	Name of State/ District	Major lac producing areas
25	Bharuch	Bharuch, Hansot, Jhagadia, Netrang and Vagra
26	Panchamahal	Dakor, Godhara, Ghoghamba, Nadiad and Santrampur
27	Vadodara	Chhota Udepur, Devhant, Jambaguda, Jatpurpavi, Jhonjh, Kawant, Keori, Kundal Ghata, Tejgadh and Zoz
VI.	**Jharkhand**	
28	Bokaro	Petarvar
29	Chatra	Tandwa
30	Dhanbad	Baghmara, Baliapur, Nirsa, Topchanchi, Tundi and Purbi Tundi
31	Deoghar	Devipur, Mohanpur, Sarwan, Sarath and Margomunda
32	Dumka	Gopikandar, Kathikund, Masaliya, Ramgarh and Shikaripara
33	East Singhbhum	Boram, Chakulia, Dhalbhumgarh, Gurabandha Musabani, Potka and Patamda
34	Giridih	Bagodar, Bengabad, Dumri, Pirtand, Suriya and Tisri
35	Garhwa	Bargarh, Garhwa, Godarmana, Rakshi, Ramganga, Ramkonda, Ranka
36	Gumla	Chainpur, Dumari, Kanshir, Palkot, Patratoli, Raidih
37	Hazaribag	Barkagaon, Barkatha, Bishnugarh, Chalkusha, Chouparan Churchu, Dadi, Daru, Ichak, Katkamdag
38	Jamtara	
39	Khunti	Karra, Murhu, Rania, Soeko Torpa, Tapkara
40	Latehar	Balumath, Barwadih Brahmani, Chandwa, Garu Latehar, Manika, Sarju, Satbarwa
41	Lohardaga	
42	Palamu	Matalong, Panki, Raj Chaipur
43	Ranchi	Angara, Banta, Birbanki, Bundu, Jonha, Namkum, Ormanjhi, Silli, Maranghada, Sonahatu, Tamar
44	Saraikela Kharsanwa	
45	Simdega	Bano, Hating hode, Jaldega, Kolebira, Lachragarh
46	West Singhbhum	Anandpur, Bandgaon, Baskata, Chaibasa, Chakradharpur, Goelkera, Lodai, Manoharpur, Sonuwa, Toklo
VII.	**Madhya Pradesh**	
47	Anuppur & Shahdol	Jaitahari, Keshwahi, Kotma, Venkatnagar
48	Balaghat	Baihar, Katangi, Lalbarra, Lamta, Langi, Parashwada, Waraseoni

Sl. No.	Name of State/ District	Major lac producing areas
49	Chhindawada	Damoa
50	Dindori	Bazak, Bhanupur, Karanjia, Ramnagar, Rampur
51	Hosangabad	Bankhedi, Babai, Daggrai, Darawpadaw, Hapa, Jonahata, Kekra, Lokamti, Pipariya
52	Mandla	Bichhia, Chabbi, Ghughari, Kalpi, Mahegaon, Mavai, Nainpur, Narainganj, Navas-Bablia
53	Narshinghpur	Chichli, Godarwara, Kalakhar, Kalyanpur, Nayakheda, Salechauka
54	Seoni	Barghat, Ghansore, Kahani, Kanewara, Keolari, Khamaria, Khari
55	Shahdol	Burhar, Jaitpur, Sohagpur
VIII.	**Maharashtra**	
56	Bhandara	Gobarwahi, Nakadongri
57	Chandrapur	Bandh, Navargaon
58	Garhchiroli	Allapalli, Bamragarh
59	Gondia	Amgaon, Chopa, Goregaon, Hirapur, Kampta, Kati, Kotjamura, Kurodhi, Salekasa, Tiroda
IX.	**Meghalaya**	
60	Garo Hills	Damra, Dodno, Nongpoh, Tura
X.	**Odisha**	
61	Balasore	Haldipada, Jaleshwar, Nilagiri
62	Keonjhar	Telkoi
63	Koraput	Ramgiri
64	Mayurbhanj	Jashipur, Kaptipada, Karanjia, Kusumi, Padampokhari, Sarat, Thakurmunda, Udala,
65	Nabarangpur	Chandahandi, Raighar
66	Sundergarh	Gurundia, Khandadhar, Kutra, Rajgangpur
XI.	**Uttar Pradesh**	
67	Sonbhadra	Doodhi
68	Allahabad	Koraon, Meja
XII.	**West Bengal**	
69	Bankura	Idpur, Khatra, Raipur, Ranibandh,
70	Midnapur	Katai, Kuti, Moyna, Panskurah, Ramnagar, Tamluk
71	Purulia	Ayodhya Pahar, Bagh Mandi, Balarampur, Jhalda, Kashipur, Kutidih, Raghunathpur, Tulin

Annexure III. Gums producing districts and areas

Sl. No.	Name of State/ District	Major gum producing areas
XIII.	**Andhra Pradesh**	
1.		Jannaram, Kamma Reddy, SK Nagar and Utnoor
2.		Chittur, Kaddapa and Nellore
3.		Addategella, Maredumilli, Rajavomma Kangi and Rampachodavaram
4.		Bhadrachalam, Chintoor, Dammapeta and Kukunaru
5.		Mannanoele, Nanjaria and Pedadornala
6.		Pathpattanam and Setampeda
7.		Arku, Chintapalli, GKVidhi, G.Murugala, Kashipatnam, Koyuru, Munchingput, Paderu and Petabayallu
8.		Gummalaxmipuram, Parvatipuram and Salur
9.		Etrunagram, Mulug and Narshimhpeda
10.		AR Puram
XIV.	**Chhattisgarh**	
11	Balrampur	Chando, Dharmi
12	Bastar	Jagdalpur, Keshkal
13	Bilaspur	Dindori, Lormi, Mugeli, Ratanpur and Takhtpur
14	Dhamtari	Gatta Silli, Nagri, Sihawa
15	Gariyaband	Amlipadar, Chhaila, Chhura Gariyaband, Indagaon, Mainpur, Udanti
16	Janjgir-Champa	Sakti, Saragaon
17	Kanker	Antagarh, Bhanupratapur, Biragaon, Kondagaon, Korar, Narharpur, Sambalpur
18	Korba	Bhaisama Bazar, Chaitama, Chhuri, Haldi Bazar, Kartala, Pali, Pasan, Pasarkhet
19	Korea	Bhartarpur
20	Mahasamund	Bagbahara, Basana, Khalari, Pithora
21	Narayanpur	Narainpur
22	Raigarh	Khamariya, Dharmjaigarh, Manendragarh Milupara, Pathalgaon
23	Raipur	Abhanpur
24	Rajnandgaon	Khardi, Manpur, Mohala-chowki
25	Sarguja	East Sarguja, South Sarguja

Sl. No.	Name of State/ District	Major gum producing areas
26	Sukma	Sukma
27	Bijapur	Bijapur, Bairamgarh, Bhopalpatnam, Usoo
28	Dantewada	Dantewada,Geedam,Kuwakonda, Katekalyan
XV.	**Gujarat**	
29	Balsar	Dharampur, Malanpada and Pangarbari
30	Banskantha	Ambaji, Pata and Virampur
31	Bharuch	Jaghadiya and Jankhawav
32	Dahod	Baria and Sagtala
33	Dang	Ahwa and Waghai
34	Kuchachh	Bhachau, Bhirandiyara, Bhuj, Dwarika, Mandvi, Nakhatrana and Rapar
35	Narmada	Dediapada, Mandvi and Rajpipla
36	Navsari	Ankalachh, Bansda and Chikhali
37	Panchamahal	Dakor, Ghoghamba, Nadiad and Santrampur
38	Sabarkantha	Bhiloda, Choriwada Modasa and Posina
39	Vadodara	Chhota Udepur, Devhant, Jambaguda, Jatpurpavi, Kawant, Tejgadh and Zoz
XVI.	**Jharkhand**	
40	Chatra	Lawalang, Pratappur and Kanti
41	Garhwa	Ramkanda and Bhandaria
42	Gumla	Dumari, Kanshir, Palkot, Patratoli, Raidih
43	Khunti	Murhu, Soeko Rania,Tapkara
44	Latehar	Garu, Mahuadar, Herhanj, Balumath, Barwadih, Lesliganj, Chhipadohar and Richughutu
45	Palamu	Daltonganj,Panki, Chhatarpur
46	Ranchi	Angara, Bundu, Jonha, Sonahatu, Tamar
47	West Singhbhum	Chakradharpur
XVII.	**Madhya Pradesh**	
48	Balaghat	Kochewahi, Baihar, Lamta, Langi, Paraswada
49	Betul	Savani
50	Chhindawada	Amarwada, Damua, Patalkot
51	Hosangabad	Pipariya
52	Mandla	Pindarai, Bamhani, Mohgaon, Lingapondi, Bichhia, Chabbi

Sl. No.	Name of State/ District	Major gum producing areas
53	Narshinghpur	Chichli, Godarwara, Kalakhar, Kalyanpur, Nayakheda, Salechauka
54	Seoni	Bhimgarh, Chhopara, Dhuma
XVIII.	**Maharashtra**	
55	Bhandara	Gobarwahi, Nakadongri
56	Chandrapur	Mul
57	Garhchiroli	Armon, Wadsa, Desai ganj
58	Gondia	Salekasha, Amgaon, Goregaon, Dhapewada, Tiroda, Navegaon Bandh
59	Wardha	Dhagabhawan, Navargaon, Hingni, Bordhara , Karanja range
XIX.	**Odisha**	
60	Balasore	Haldipada, Jaleshwar, Nilagiri
61	Keonjhar	Telkoi
62	Koraput	Ramgiri
63	Mayurbhanj	Jashipur, Kaptipada, Karanjia, Kusumi, Padampokhari, Sarat, Thakurmunda, Udala,
64	Nabarangpur	Chandahandi, Raighar
65	Sundergarh	Gurundia, Khandadhar, Kutra, Rajgangpur
XX.	**Rajasthan**	
66	Ajmer	Beawar, Kekeri, Kisangarh, Nasserabad, Puskar
67	Alwar	Bansur, Rajgarh, Ramgarh,Thanagazi
68	Churu	Sardarsahar, Sujangarh,Taranagar
69	Jaipur	Amer, Bassi, Chomu, Chaksu, Jamwa Ramgarh, Kotputli, Phagi, Phulera
70	Jhunjhunu	Chirawa, Khetri, Navalgarh
71	Jodhpur	Bilara, Luni, Phalodi, Shergarh
72	Nagour	Ladnu, Merta, Parbatsar
73	Pali	Bali, Sojat, Sumerpur
74	Sikar	Fatehpur, Neem-ka-Thana, Ringus, Sri Madhopur
75	Tonk	Niwai, Malpura, Uniara
76	Udaipur	Gogunda, Kotra

Annexure IV. Legal categorization of major gums produced in various states of India

State	MFP Under State Monopoly
Andhra Pradesh	Bamboo, tendu leaves and 25 other MFPs.
Chhattisgarh	Bamboo, tendu leaves, sal seed, myrobalans and gums of kullu, dhawda, khair and babool.
Gujarat	Bamboo, tendu leaves, mhowra flower, mhowra seed, gum karaya and other miscellaneous gums.
Maharashtra	Bamboo, tendu leaves, apta leaves and 33 other MFPs.
Madhya Pradesh	Bamboo, tendu leaves, sal seed, kullu gum and lac resin.
Rajasthan	Bamboo, tendu. State monopoly for other MFPs ceased to exist in 2003.
Odisha	Bamboo and tendu leaves.
Himachal Pradesh	No MFP has been put under State monopoly.
Jharkhand	Tendu leaves.

Source: *TRIFED, Ministry of Tribal Affairs, New Delhi*

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